The role of overconfidence in entrepreneurial exit decisions: linking two distinct streams of literature

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List of Abbreviations

IPO	Initial public offering
M&A	Mergers & acquisition
NPV	Net present value

1 Introduction and Motivation

Entrepreneurial activity is inherently time-bound. New ventures are created in response to recognized opportunities, enter the market, expand over time, and are eventually relinquished. While much of entrepreneurship research focuses on opportunity recognition, market entry, and venture growth, the decision to exit a venture is equally fundamental and deserves systematic investigation. Exit is not an exceptional or residual outcome but a core component of entrepreneurial decision-making (DeTienne, 2010).

Importantly, exit decisions are rarely discrete or made at a single point in time. Entrepreneurs typically face repeated decisions about whether to continue with the current course of action, adapt their strategy, or disengage from their ventures under conditions of uncertainty and incomplete information (DeTienne, 2010; Dixit & Pindyck, 1994). Understanding when and how entrepreneurs exit is therefore essential for a comprehensive understanding of the entrepreneurial process as a whole.

1.1 Research Problem and Relevance

"Despite the fact that all entrepreneurs will eventually exit their firms, very little research exists to document or explain this phenomenon" (DeTienne & Cardon, 2012, p. 351).

Entrepreneurial exit is an inherent element of the entrepreneurial process. Founders will eventually disengage from the ventures they create. Yet both the timing and the mode of exit vary substantially across individuals and contexts (DeTienne, 2010; DeTienne et al., 2015). While early entrepreneurship research focused on entry and growth, exit received comparatively limited attention. More recent work conceptualizes exit as a forward-looking decision process (DeTienne, 2010), which Chapter 2 reviews in more detail. (Dixit & Pindyck, 1994; McDonald & Siegel, 1986)

Despite these developments, an important empirical regularity remains insufficiently explained. Entrepreneurs frequently delay exit even in situations when objectively observable indicators point to persistent underperformance (Sandri et al., 2010). From a static net present value (NPV) perspective, continued operation under negative expected returns would imply exit. Real options theory refines this benchmark by recognizing the value of waiting under uncertainty (Dixit & Pindyck, 1994). However, a growing body of empirical and experimental evidence suggests that observed exit delay cannot be fully explained by real options logic alone (Sandri et al., 2010; Schade & Snir, 2020).

This systematic heterogeneity in exit timing raises a central theoretical question: Why do entrepreneurs facing similar performance trajectories and environments arrive at different exit decisions? Existing explanations emphasize biases such as escalation of commitment or status quo bias (Samuelson & Zeckhauser, 1988; Staw, 1981). While these capture important aspects of persistence, they are largely backward-looking, i.e., they attribute persistence primarily to past investments and prior commitments. Hence, they do not fully account for heterogeneity in forward-looking expectations under uncertainty.

Schade and Snir's (2020) decision-not-to-decide framework provides a critical conceptual shift. It reframes exit delay as a choice over the timing of irreversible commitment rather than as passive inertia or error. Their framework shows that postponement can arise as a systematic behavioral decision outcome under uncertainty. At the same time, the framework does not specify how subjective beliefs about future outcomes shape the perceived attractiveness of postponement compared to immediate exit.

This thesis advances the argument that explanations of heterogeneity in exit timing remain incomplete without explicitly modeling belief distortions. It adopts a behavioral real options perspective in which exit delay reflects objective option value as well as subjective evaluations of continuation under uncertainty. Specifically, it integrates overconfidence into the decision-not-to-decide framework of Schade and Snir (2020). By systematically biasing expectations about venture prospects, relative performance, and forecast precision, overconfidence provides a belief-based mechanism through which postponement becomes attractive and exit thresholds diverge across entrepreneurs (Hayward et al., 2006; Moore & Healy, 2008). This suggests that exit timing cannot be fully understood without analyzing how entrepreneurs form and revise expectations under uncertainty.

1.2 Research Objective

The objective of this thesis is to analyze how overconfidence influences irreversible entrepreneurial exit decisions under uncertainty. Specifically, it examines the effects of overconfidence on both exit timing and exit mode choice. These effects are examined using the decision-not-to-decide framework proposed by Schade and Snir (2020).

Instead of treating exit as a binary outcome, this thesis conceptualizes exit as a dynamic disinvestment decision. Entrepreneurs repeatedly choose between committing to immediate exit and postponing irreversible commitment while remaining active in the venture (Wennberg & DeTienne, 2014).

The analysis focuses on three distinct forms of overconfidence: overestimation, overplacement, and overprecision. These forms bias how entrepreneurs evaluate continuation relative to exit. In particular, they can shift subjective exit thresholds, inflate the perceived option value of waiting, and influence the evaluation of external exit options, e.g., sales or acquisitions (Moore & Healy, 2008).

The main research question is:

How does overconfidence influence entrepreneurial exit timing and exit mode choice under uncertainty, and how can these effects be theoretically explained within the decision-not-to-decide framework of Schade and Snir (2020)?

This question is addressed through the following sub-questions, which map directly onto the propositions developed in Chapter 5:

- How does overestimation affect exit timing, particularly the likelihood of delayed exit from underperforming ventures (P1)?
- How does overplacement distort the evaluation and acceptance of external exit options, thereby shaping exit mode choice and, in many cases, delaying exit (P2, P5)?
- How does overprecision influence belief updating and generate exit inertia when negative performance signals accumulate (P3)?
- How does overconfidence increase the adoption of a decision-not-to-decide strategy beyond what real options logic alone would predict (P4)?
- How do individual characteristics, specifically gender and age, moderate the relationship between overconfidence (and its specific forms) and exit delay or exit inertia (P6, P7)?

By explicitly distinguishing between mechanisms that affect exit timing and those that affect exit mode choice, this thesis provides a structured account of how belief formation and belief updating shape entrepreneurial disengagement.

1.3 Intended Contribution (Theoretical & Empirical)

This thesis makes three primary theoretical contributions:

First, it contributes to the entrepreneurial exit literature by offering a belief-based explanation for heterogeneity in exit timing. This explanation goes beyond objective venture performance and the option value of waiting predicted by real options theory. By integrating overconfidence into a

dynamic exit framework, this thesis shows how subjective beliefs systematically shift exit thresholds and sustain postponement.

Second, it extends the decision-not-to-decide framework by embedding it within a behavioral real options perspective. While Schade and Snir (2020) show that postponement can arise as a coping response to irreversibility and decisional stress, this thesis explains why postponement becomes persistently attractive. It does so by introducing biased belief formation as a complementary explanatory mechanism.

Third, this thesis contributes to the overconfidence literature by moving beyond entry and persistence to analyze entrepreneurial exit decisions. It shows how different forms of overconfidence affect distinct dimensions of exit behavior.

Although the contribution is primarily theoretical, the proposed framework yields clear and testable propositions. These propositions provide a basis for future empirical and experimental research.

1.4 Structure of this Thesis

This thesis is structured as follows:

Chapter 2 reviews the literature on entrepreneurial exit. It introduces exit typologies and conceptualizes exit timing as a dynamic decision under uncertainty. Chapter 3 develops a conceptual framework of overconfidence in entrepreneurship. It distinguishes between overestimation, overplacement, and overprecision and outlines their behavioral implications.

Chapter 4 introduces the decision-not-to-decide framework of Schade and Snir (2020). It integrates their framework with a behavioral real options perspective and specifies the decision-theoretic structure within which overconfidence operates. Chapter 5 develops a set of propositions that link distinct forms of overconfidence to exit timing, exit mode choice and exit inertia.

Chapter 6 discusses the theoretical and practical implications of the proposed framework, while Chapter 7 concludes by summarizing the key insights and stating the overall contribution of this thesis.

2 Theoretical Background I: Entrepreneurial Exit

This chapter establishes the theoretical foundations of entrepreneurial exit. It defines the concept of entrepreneurial exit, introduces exit typologies, and reviews key determinants of exit timing at the individual, venture, and environmental levels.

2.1 Conceptual Foundations of Entrepreneurial Exit

This section clarifies the conceptual scope of entrepreneurial exit, discusses its relevance, and introduces typologies that capture heterogeneity in exit outcomes.

2.1.1 Entrepreneurial Exit: Definition and Importance

Entrepreneurial exit is a fundamental part of the entrepreneurial lifecycle, yet for a long time it received limited attention in entrepreneurship research. Early studies focused mainly on opportunity recognition, venture creation, market entry, or firm growth. Exit was often treated as a residual or final outcome of the entrepreneurial process rather than as a phenomenon on its own (DeTienne, 2010). Peters (2009) similarly observed that exits remain the least understood part of investing. Entrepreneurs and investors tend to focus on starting, scaling, and growing ventures.

An entrepreneurial exit occurs when founders leave ventures they have helped to create. Entrepreneurial involvement is therefore inherently time-bound. The relevant question is not whether entrepreneurs exit but through which modes, under which conditions, and with which consequences. In any case, this moment represents a critical transition in the entrepreneurial process. Exit involves the reallocation of ownership and decision-making rights as well as entrepreneurial effort. As a consequence, exit affects both the future of the firm and the founder (DeTienne & Cardon, 2012).

DeTienne (2010) defines entrepreneurial exits as “the process by which the founders of privately held firms leave the firm they helped to create; thereby removing themselves, in some degree, from the primary ownership and decision-making structure of the firm” (p. 204). By framing exit as a process, DeTienne (2010) clarifies that entrepreneurial exits unfold over time rather than occur as a single, discrete event. This definition provides the conceptual foundation for the present thesis.

Entrepreneurial exit must be distinguished from business failure or firm death. Exit does not necessarily imply poor performance or failure, as entrepreneurs may leave their ventures under both favorable and unfavorable conditions (DeTienne, 2010; Headd, 2003; Wennberg et al., 2010). Accordingly, exits encompass a wide range of outcomes in which founders leave their ventures. These outcomes include financially successful exits, e.g., initial public offerings (IPOs) or mergers

and acquisitions (M&As), as well as exit driven by non-financial motives or constraints (DeTienne & Cardon, 2012; DeTienne et al., 2015). Earlier research treated exits in overly simplistic terms, portraying them as binary outcomes. Exit signaled failure and survival signaled success (Wennberg & DeTienne, 2014).

In recent years, entrepreneurial exits have become increasingly relevant in academic research and entrepreneurial practice. Exit strategies and their underlying reasons as well as the timing of exit have become central topics (DeTienne, 2010). One reason is the large financial impact of exit events. For founders, exits often represent the primary opportunity to realize financial returns from years of investment, entrepreneurial effort, and risk-taking. For investors, e.g., venture capitalists, exits are the central mechanism through which returns are generated (Cumming & MacIntosh, 2003). Industry reports, including those by PricewaterhouseCoopers, underscore the macroeconomic importance of exit markets. They emphasize the role of capital reallocation and economic growth (PricewaterhouseCoopers, 2025).

Over the past decade, researchers have begun to integrate exit more explicitly into theories of the entrepreneurial process. Rather than treating exit as an endpoint, recent research increasingly conceptualizes entry, persistence, growth, and exit as interdependent stages of a dynamic process (DeTienne, 2010; Wennberg & DeTienne, 2014). From this perspective, exit decisions can be analyzed though the lens of real options theory. Entrepreneurs compare the expected continuation value of the venture with the value of exit under uncertainty (Dixit & Pindyck, 1994).

2.1.2 Entrepreneurial Exit Typologies

To account for the heterogeneity of entrepreneurial exit outcomes, the literature distinguishes between different types of exits. Instead of treating exit as a singular occurrence, contemporary research points out that entrepreneurs exit for different reasons and through different mechanisms (DeTienne, 2010; DeTienne & Cardon, 2012).

DeTienne et al. (2015) synthesize prior research and condense entrepreneurial exits into a typology of three higher-order exit strategies: financial harvest, stewardship, and voluntary cessation. They also empirically test predictors of exit-type choice. These exit strategies differentiate between different modes through which entrepreneurs intend to leave their venture. They differ in their observable outcome as well as in their underlying motivation, timing, and decision-making processes.

2.1.2.1 Harvest Exit

Financial harvest exits refer to an exit mode in which entrepreneurs primarily aim to maximize financial returns. In this case, the venture typically continues to operate as an independent entity, though often under new ownership and governance structures. Entrepreneurs relinquish ownership and control either partially or fully. Common examples include the sale of the firm through IPOs or M&As (DeTienne et al., 2015).

Harvest exits are commonly the outcome of forward-looking strategic planning. Entrepreneurs pursuing this exit strategy often adopt a desire for growth early on, e.g., scaling aggressively or hiring a professional management team. Empirically, DeTienne et al. (2015) show that harvest exit strategies are strongly associated with the entrepreneur's perception of high innovativeness and the use of causation-based decision-making processes. Consequently, harvest exits are frequently linked to comparatively strong venture performance and high growth potential.

At the same time, pursuing a financial harvest exit involves a trade-off between financial returns and control. While financial harvest exits allow entrepreneurs to monetize their ventures, they also require relinquishing strategic control and, in many cases, disengaging from the firm entirely. Thus, not all successful ventures pursue harvest exits. In terms of timing, harvest exits are typically opportunity driven. Entrepreneurs seek to exit when valuations are high or when external market conditions, e.g., acquisition demand, are particularly favorable (DeTienne et al., 2015).

2.1.2.2 Stewardship Exit

Stewardship exits occur when founders transfer ownership rights and control of the firm. The primary objective is preserving the venture's identity and core values. In contrast to financial harvest exits, financial returns are often secondary to non-financial motives, e.g., legacy, family continuity, employee welfare, or social responsibility. Typical examples include succession within the family, employee buyouts, or sale of the firm to an individual who will then personally manage and continue the business (DeTienne et al., 2015).

Entrepreneurs who are pursuing a stewardship exit tend to exhibit pro-social and pro-organizational behavior, consistent with stewardship theory (Davis et al., 1997). DeTienne et al. (2015) find that stewardship exit strategies are negatively associated with extrinsic (financial) motivation and positively associated with a strong motivation for autonomy. Founders seek to retain discretion over exit decisions in order to ensure that the firm is transferred to successors who are aligned with its values.

Exit decisions in this mode are consequently often shaped by emotional attachment and identity considerations. As continuity is prioritized over immediate financial gains, stewardship exits are frequently associated with later exit timing and prolonged decision-making processes. Entrepreneurs may delay an exit until a suitable successor is identified or until organizational and personal conditions align with their long-term objectives. Accordingly, stewardship exits are less sensitive to short-term performance fluctuations and more influenced by relational and normative considerations (DeTienne et al., 2015).

2.1.2.3 Voluntary Cessation

Voluntary cessation refers to an exit mode in which entrepreneurs terminate the business activity altogether. The venture does not continue to operate after exit (DeTienne et al., 2015). Consistent with earlier discussion, voluntary cessation should not be equated with entrepreneurial failure, as exit does not necessarily imply that the venture or the entrepreneur has failed (Headd, 2003).

Voluntary cessation may result from persistent underperformance or financial distress. However, it can also reflect strategic decisions. Entrepreneurs may choose to exit when the primary activity of the venture ends or when the firm fulfills the purpose for which it was created. In such cases, exit reflects a conscious disengagement from the venture instead of an involuntary collapse (DeTienne et al., 2015).

Within this typology, DeTienne et al. (2015) differentiate between different forms of closure. Liquidation refers to the process of selling off a venture's assets in order to repay creditors and distribute any remaining value. Discontinuance involves simply ceasing operations without a formal asset liquidation process. Empirically, voluntary cessation exits are associated with low perceived innovativeness, few employees, and limited use of formal planning and causation-based decision-making. Although such exits are often linked to weaker economic outcomes at the firm level, they do not necessarily imply negative outcomes for entrepreneurs. Prior research indicates that voluntary cessation can free up resources and enable entrepreneurs to pursue new opportunities (DeTienne et al., 2015; Headd, 2003).

2.2 Exit Timing and Determinants

This section conceptualizes entrepreneurial exit as a dynamic timing decision under uncertainty and reviews determinants of exit timing across the individual, venture, and environmental levels.

2.2.1 Exit Timing as a Dynamic Decision under Uncertainty

Entrepreneurial exit is best conceptualized as a timing problem rather than a binary choice between continuation and termination (Wennberg & DeTienne, 2014). From this perspective, exit

resembles a dynamic capital allocation decision in which entrepreneurs repeatedly reassess whether continued commitment yields greater expected value than reallocating resources to alternative uses (Dixit & Pindyck, 1994). The central question therefore is not only whether to exit but when exit becomes optimal under uncertainty.

Exit timing decisions are made under conditions of uncertainty, irreversibility, and gradually unfolding information (Dixit & Pindyck, 1994). Exit timing is better understood as a decision under uncertainty rather than measurable risk, as entrepreneurs typically lack reliable probability distributions for future outcomes. Uncertainty arises because venture prospects, e.g., venture performance, future demand, competitive dynamics, or technological developments, cannot be predicted reliably. At the same time, exit is often partly irreversible. Leaving the venture can entail sunk costs and foregone upside potential. Once entrepreneurs exit, re-entry is often very costly or infeasible. Because information about venture prospects arrives gradually over time, entrepreneurs can in principle learn and revise expectations. This makes the timing of exit a forward-looking decision (DeTienne, 2010; Dixit & Pindyck, 1994).

Real options theory provides a useful framework for exit timing under these conditions. Instead of treating exit as an immediate response to current performance, real options logic emphasizes the value of flexibility. Continued operation preserves the option to wait and benefit from favorable future developments. Under certain conditions, this option value can make delayed exit optimal (Dixit & Pindyck, 1994). In contrast, exiting terminates this option and fixes outcomes at their current level.

This distinction also clarifies why a purely static NPV approach can be descriptively limited in irreversible decision settings. A myopic NPV rule treats negative NPV as a sufficient condition for termination and ignores the value of waiting, which can be substantial under uncertainty (Dixit & Pindyck, 1994; McDonald & Siegel, 1986).

Nevertheless, empirical and experimental evidence suggests that observed exit delay often exceeds what real options value considerations alone would predict. Using a controlled experimental setting, Sandri et al. (2010) show that participants frequently postpone termination even when exit would dominate continuation under an optimal-stopping benchmark. A substantial share of decision-makers exhibit inertia that is only weakly related to rational disinvestment triggers and inconsistent with a simple NPV rule. Notably, entrepreneurs and non-entrepreneurs behave largely similarly in this setting. These findings indicate that delayed exit reflects not only uncertainty-

driven incentives to wait but also behavioral frictions that dampen responsiveness to negative information and sustain continuation beyond economically optimal points (Sandri et al., 2010).

Additionally, a useful implication of this evidence is to distinguish between two conceptually different sources of delay. Options-based inertia refers to postponement that can be justified by rationally considering the value of waiting (Dixit & Pindyck, 1994). By contrast, psychological inertia refers to persistence that remains after accounting for rational option value (Sandri et al., 2010). This distinction matters because it implies that heterogeneity in exit timing cannot be explained by objective venture conditions alone.

Taken together, exit timing reflects a trade-off between the option value of waiting and the costs of continued commitment. Entrepreneurs must weigh flexibility and learning opportunities against the risk of accumulating further losses as venture prospects deteriorate (Dixit & Pindyck, 1994). The evidence that the timing of exit often exceeds real options benchmarks motivates closer attention to judgment and belief-based mechanisms, i.e., how entrepreneurs interpret signals, form expectations, and evaluate continuation relative to exit under uncertainty.

2.2.1.1 Uncertainty

Uncertainty complicates the timing of exit by affecting the relative value of waiting versus exiting (Dixit & Pindyck, 1994). Under real options logic, continued operation preserves an option value. By delaying exit, entrepreneurs retain exposure to favorable future developments, e.g., demand growth or technological change. In contrast, exiting fixes payoffs at their current level and eliminates the possibility for future upside.

In financial terms, delaying exit preserves flexibility but also maintains exposure to downside risk. Accordingly, entrepreneurs face a trade-off between the value of waiting and the risk of future losses. This makes exit timing a trade-off between downside exposure and upside optionality instead of a response to current performance (Dixit & Pindyck, 1994).

Within standard real options logic, higher levels of uncertainty increase the value of waiting. This can make delayed exit optimal even when current performance is weak. Persistence and exit delay may consequently reflect rational responses to unresolved uncertainty (Dixit & Pindyck, 1994).

2.2.1.2 Irreversibility

Another key characteristic of exit decisions is irreversibility. Once entrepreneurs exit a venture, re-entry is often very costly or infeasible (Dixit & Pindyck, 1994). Irreversibility may arise from sunk investments, which Dixit and Pindyck (1994) describe as investments that are firm- or industry-specific and cannot be recovered, e.g., most expenditures on marketing and advertising.

Irreversibility implies that exit decisions involve the potential destruction or realization of remaining firm value. Once costs are sunk, rational continuation should depend on expected NPV and not the sum of past investments alone. However, irreversibility often makes it difficult for entrepreneurs to disregard prior commitments when evaluating exit timing (Dixit & Pindyck, 1994).

Moreover, irreversibility creates a trade-off. Exiting too early may destroy future value if conditions improve, whereas delaying exit may prolong losses if venture prospects continue to deteriorate (Dixit & Pindyck, 1994).

2.2.1.3 Information and Learning

Exit decisions are shaped by the gradual resolution of uncertainty as new information becomes available. Entrepreneurs continuously update their beliefs about venture prospects. Real options theory emphasizes this learning process and suggests that optimal timing depends not only on the level of uncertainty but also on how quickly uncertainty resolves and how valuable and informative new signals are (Dixit & Pindyck, 1994).

Further, learning under uncertainty involves both the arrival of objective information and the subjective interpretation of that information (Dixit & Pindyck, 1994). While entrepreneurs may be exposed to similar signals, they can differ substantially in how strongly they interpret performance signals and revise expectations about future cash flows. Learning therefore does not translate mechanically into action but is mediated by individual judgments and interpretations. As a consequence, entrepreneurs facing comparable objective conditions may assign different continuation values to the same venture. This is associated with persistent heterogeneity in exit timing (Wennberg & DeTienne, 2014).

Consistent with the distinction between option-based and psychological inertia discussed above, persistence in disinvestment can occur even when additional learning value is low. This can result in delayed exit beyond real options benchmarks (Sandri et al., 2010).

Persistence in underperforming ventures has been demonstrated to reflect not only expectations about future recovery but has been shown to depend partly on motivational forces that sustain commitment even in scenarios where economic indicators deteriorate (DeTienne et al., 2008). Exit decisions are therefore inherently forward-looking (DeTienne, 2010). Entrepreneurs may tolerate weak current performance if they expect future information to clarify venture prospects or reveal recovery opportunities. Differences in expectations and interpretations of uncertainty can therefore lead to divergent exit behavior (Gimeno et al., 1997; Wennberg & DeTienne, 2014).

Moreover, belief updating is not a purely mechanical process. Entrepreneurs may exhibit cognitive biases, e.g., overconfidence, which systematically affect how new information is processed and integrated into exit decisions (Hayward et al., 2006). These can lead entrepreneurs to delay exiting beyond economically optimal points or to exit prematurely despite positive option value. Observed exit timing may deviate from predictions derived from standard real options models. Such escalation dynamics are consistent with the broader literature on escalation of commitment, which show that decision-makers often continue investing in failing courses of action to justify prior investments and avoid acknowledging losses (Staw, 1981).

Additionally, a related behavioral pattern has been documented in financial decision-making. Investors tend to hold on to losing positions for too long while selling winning positions too early. This phenomenon is known as the "disposition effect". This bias reflects loss aversion and a reluctance to realize losses, as selling at a loss forces individuals to acknowledge negative outcomes and foregone expectations (Shefrin & Statman, 1985). In entrepreneurial contexts, a similar mechanism may operate when founders delay exit decisions in order to avoid realizing losses or revising prior success expectations (Gimeno et al., 1997; Shepherd, 2003). Although entrepreneurial exit differs from asset trading, both settings involve irreversible decisions under uncertainty and psychological costs of loss realization (Dixit & Pindyck, 1994).

Taken together, empirical evidence suggests that exit timing reflects a combination of rational learning under uncertainty and behavioral biases that dampen responsiveness to negative information. Consistent with Sandri et al. (2010), delayed exit cannot be explained by uncertainty or option value alone but reflects persistent behavioral frictions in belief updating and disengagement decisions.

2.2.1.4 Exit as a Disinvestment Decision

Entrepreneurial exit can be conceptualized as a disinvestment decision. This decision mirrors the initial investment choice and reflects the structural symmetry between entry and exit under uncertainty. From this view, exit corresponds to the termination of an ongoing investment project. It is subject to the same economic forces of irreversibility and timing that characterize investment decisions generally. Real options theory emphasizes that abandonment is not optimal as soon as expected returns turn negative. Instead, exit becomes optimal only when the expected continuation value falls sufficiently below the value of exit, once the loss of future flexibility is taken into account (Dixit & Pindyck, 1994).

Viewing exit as a disinvestment highlights why delayed exit may occur (Dixit & Pindyck, 1994). Consequently, optimal exit timing depends not only on current performance but also on expectations about future outcomes and the relative value of maintaining flexibility.

At the same time, the disinvestment perspective clarifies the asymmetry between continuation and exit. While continued operation allows entrepreneurs to revise decisions as new information arrives, exit represents a final and largely irreversible commitment. As a result, exit thresholds tend to be higher than continuation thresholds, which contributes to inertia and persistence. This asymmetry helps explain why entrepreneurs may remain invested longer than suggested by NPV calculations (Dixit & Pindyck, 1994).

Framing exit as a disinvestment thus provides a conceptual link between rational option-based delay and observed behavioral inertia. While real options theory predicts delayed exit when flexibility has value, empirical evidence suggests that entrepreneurs may persist even when option-value considerations alone would imply exit. This tension underscores the importance of incorporating behavioral considerations into models of entrepreneurial exit timing (Sandri et al., 2010; Shepherd, 2003).

2.2.2 Individual-Level Determinants

Entrepreneurial exit timing is strongly influenced by individual characteristics that shape how entrepreneurs interpret performance signals, form expectations about venture prospects, and define subjective thresholds for acceptable continuation. Research shows that exit decisions depend not only on economic performance but also on the entrepreneurs' human capital, motivations, risk-tolerance, or identity.

Human capital is one of the most studied individual-level determinants. Education, work experience, and entrepreneurial expertise as well as alternative employment opportunities and switching costs to other occupations affect opportunity costs and flexibility. High opportunity costs imply that continued investment in a low-performing venture becomes financially less attractive. Under these circumstances, the likelihood of exit increases when expected returns fall below a subjective threshold of performance. In contrast, entrepreneurs with firm-specific human capital may delay exit due to higher switching costs (Gimeno et al., 1997).

In addition, age and career stage play an important role in exit decisions. Younger entrepreneurs typically face longer remaining career horizons and lower costs of re-entry, which can make an exit a more attractive option. By contrast, older entrepreneurs may delay exit due to more limited

alternative employment opportunities, stronger emotional attachment to their venture, or retirement-related considerations (DeTienne & Cardon, 2012; Wennberg et al., 2010).

Age may also influence exit decisions through cognitive and identity-related mechanisms. Longer professional histories and accumulated experience can reinforce established mental models and increase attachment to prior expectations, potentially reducing flexibility in belief revision (Dane, 2010). Such experience-based reasoning may shape responsiveness to disconfirming performance signals. It may also influence the speed with which entrepreneurs adjust expectations under uncertainty. Together, these effects shape the persistence of continuation decisions.

Importantly, empirical evidence on the role of age is mixed. DeTienne and Cardon (2012) show that age does not uniformly predict exit behavior across exit modes, which suggests that its effects depend on career stage and contextual factors. Consistent with this view, Gimeno et al. (1997) find that older entrepreneurs are willing to accept lower performance thresholds before exiting, reflecting higher exit costs and reduced outside options.

Furthermore, motivations and goals shape exit timing and their intended exit strategies. Entrepreneurs differ in their preferences, implying they place different importance on financial returns, autonomy, personal fulfillment, or legacy. Those with strong financial motivations are more likely to pursue earlier and financially oriented exits. Those with non-financial or intrinsic motivations may persist longer (DeTienne & Cardon, 2012; DeTienne et al., 2015; Gimeno et al., 1997).

Further, individual differences in risk tolerance and risk perception shape exit timing. Entrepreneurs vary in their willingness to bear downside risk and uncertainty, which affects how long they persist in ventures with volatile or declining performance (Gimeno et al., 1997). More risk-tolerant entrepreneurs may continue operating despite unfavorable signals, placing greater weight on potential upside outcomes, while more risk-averse individuals may prefer earlier exit to limit downside exposure.

Moreover, related literature identifies gender as one systematic source of heterogeneity in risk perceptions and responses to uncertainty. On average, female and male entrepreneurs differ in how they assess and tolerate risk, which can translate into differences in exit timing and exit thresholds (Wennberg et al., 2010). Importantly, these differences reflect subjective perceptions of risk rather than objective risk exposure, as entrepreneurial decisions are typically made under conditions of uncertainty.

Beyond differences in risk tolerance, research also documents systematic gender differences in confidence calibration and belief updating under uncertainty. On average, men tend to report higher subjective success probabilities and exhibit stronger overconfidence particularly in competitive and uncertain environments. By contrast, women's expectations are often more conservatively calibrated to objective benchmarks (Koellinger et al., 2013; Moore & Healy, 2008). Such differences in belief formation may shape how performance signals are interpreted and how quickly exit thresholds adjust in response to negative feedback.

Moreover, emotional attachment and identity also influence exit behavior. For some founders, their ventures are closely tied to personal identity and social status (DeTienne et al., 2015; Shepherd, 2003). Strong emotional attachment to their venture can increase the psychological costs of exit and raise the subjective threshold at which exit becomes acceptable (Gimeno et al., 1997). In such cases, entrepreneurs may stay beyond the economically optimal point. Business failure or exit can therefore be experienced as a personal loss that can trigger grief responses. This phenomenon may further consume emotional and cognitive resources, thereby delaying disengagement from the venture (Shepherd, 2003).

Additionally, opportunity costs vary substantially across individuals. Those with strong outside employment opportunities or broad skill sets face lower relative costs of exit. Conversely, entrepreneurs with high venture-specific human capital or an identity that is closely tied to the venture may perceive exit as more costly (Gimeno et al., 1997; Shepherd, 2003). This heterogeneity contributes to variation in exit timing among entrepreneurs facing similar venture-level conditions.

Taken together, individual-level characteristics influence exit timing not only through economic constraints or preferences but also by shaping how entrepreneurs interpret information, calibrate beliefs, and revise expectations under uncertainty. As a result, heterogeneity in exit timing among entrepreneurs facing similar venture-level conditions can be traced, in part, to systematic differences in belief formation and belief updating processes. This observation suggests that individual characteristics may not only directly affect exit timing but also condition how cognitive biases translate into persistence or disengagement.

2.2.3 Venture-Level Determinants (Micro Factors)

Venture-level determinants, or micro factors, refer to characteristics of the venture that shape its likelihood of exit and survival. A central insight from the literature is that firm performance plays a key role in exit decisions. Lower levels of performance, e.g., poor profitability and weak growth,

increase the likelihood of exit (Cefis et al., 2022). Declining performance reduces expected future returns and increases the likelihood that continuation is no longer economically viable. Entrepreneurs respond not only to current performance levels but also to the expectations of future performance. Pessimistic forecasts may trigger exit even when current performance remains acceptable (Gimeno et al., 1997; Wennberg et al., 2010).

Additionally, growth opportunities and future potential influence the timing of exit. Ventures with innovative capabilities or scalable business models tend to persist longer, as entrepreneurs delay exit in anticipation of higher future returns (Cefis et al., 2022; DeTienne et al., 2015). Conversely, limited growth potential can accelerate exit decisions, especially when alternative opportunities appear more attractive (Wennberg et al., 2010). Innovation plays a dual role in this context. While it may enhance survival by improving competitiveness, it can also facilitate exit through value-creating routes (Cefis et al., 2022).

Further, financial constraints and resource availability shape the timing of exit. Liquidity shortages or difficulties in securing external financing can precipitate early exit, even when long-term venture prospects are uncertain rather than clearly negative. In contrast, financial slack or patient capital can allow entrepreneurs to continue operating and postpone exit decisions (Cefis et al., 2022).

Moreover, ownership structure influences exit timing through its implications for control and governance. Ventures with external investors, particularly those backed by venture capital, are often subject to formal performance benchmarks or predefined exit horizons. These arrangements reduce founders' discretion over exit decisions. They may also accelerate or formalize exit timing relative to founder-controlled firms, particularly when investors face liquidity pressures or when fund lifecycle constraints become more binding (Cumming & MacIntosh, 2003; DeTienne & Cardon, 2012).

Despite the importance of these micro factors, substantial heterogeneity in exit timing remains. Firms with similar observable characteristics operating under comparable conditions often exhibit different exit behaviors. This suggests that venture-level characteristics alone are insufficient to fully explain exit timing.

In addition, exit timing is influenced by strategic interaction with external actors. Entrepreneurs may delay or accelerate exit in anticipation of competitors' actions, potential acquisition interest, or investor responses (Cefis et al., 2022; Wennberg & DeTienne, 2014). For example, founders may persist through temporary losses to signal commitment or bargaining strength, while investors

may exert pressure for earlier exit to realize liquidity. Exit timing thus reflects not only internal venture conditions but also strategic considerations regarding how exit decisions are perceived by others (Cefis et al., 2022).

2.2.4 Environmental-Level Determinants (Macro Factors)

Environmental-level determinants, or macro factors, also play a crucial role in exit timing. These factors capture features of the external environment that influence firms' survival prospects and entrepreneurs' expectations about future performance. Macro factors, e.g., technological change, demand volatility, and shifts in competition, increase uncertainty and can complicate forecasts of future returns (Cefis et al., 2022; Wennberg & DeTienne, 2014). Increased uncertainty can therefore affect exit decisions not only by deteriorating current performance but also by altering expectations about future viability (Wennberg & DeTienne, 2014). From a real options perspective, environmental uncertainty can increase the value of waiting and delay exit by amplifying the costs of irreversible commitment (Dixit & Pindyck, 1994).

Further, industry structure and competition strongly affect exit timing by influencing expectations of future profitability. In highly competitive industries with low entry barriers and intense price pressures, firms see limited chances of recovery and may choose to exit earlier. In contrast, high entry barriers and temporary advantages, e.g., patent protection, shield firms from competitive pressure. These conditions allow entrepreneurs to anticipate future profits and withstand competitive pressure longer (Cefis et al., 2022).

Additionally, regulatory and institutional conditions influence the timing of entrepreneurial exit. Changes in regulation, taxation, labor laws, or environmental standards can alter the costs and benefits of continuing operations. Increased regulatory burdens may accelerate exit by raising operating costs or uncertainty. On the other hand, deregulation or supportive policy environments can shape exit decisions and the viability of continued operation by influencing expected payoffs (Wennberg & DeTienne, 2014).

Moreover, exit market conditions represent an additional environmental determinant of exit timing. The availability and attractiveness of exit modes directly affect the timing and feasibility of exit. Favorable acquisition markets or high valuations may encourage earlier exit. Entrepreneurs may seek to exit to capitalize on attractive opportunities. In contrast, weak exit markets may delay exit, as entrepreneurs wait for more favorable conditions before realizing value (Cumming & MacIntosh, 2003; Wennberg & DeTienne, 2014).

Environmental determinants interact with individual- and venture-level characteristics. Similar macroeconomic shocks or market conditions may lead to different exit responses depending on entrepreneurs' financial positions and interpretations of venture prospects (Wennberg & DeTienne, 2014). For example, heightened uncertainty may delay exit for entrepreneurs with sufficient financial slack, while accelerating exit for those facing liquidity constraints. Likewise, favorable exit markets may prompt early exit for financially motivated founders but exert less influence on entrepreneurs pursuing non-financial exit goals. These interacting effects imply that exit timing cannot be explained by linear relationships between isolated factors. Rather, it reflects the joint influence of environmental and venture-specific conditions and individual decision-making processes.

2.3 Research Gap

This section identifies a gap in existing literature. It argues that objective economic determinants alone cannot fully explain and predict heterogeneity in exit timing, which motivates a behavioral perspective on entrepreneurial exit decisions.

Even when ventures follow comparable performance trajectories and operate in similar environments, founders often choose different exit timings and exit modes. Experimental evidence further shows systematic persistence in disinvestment decisions. Individuals tend to hold on to projects longer than implied by real options benchmarks (Sandri et al., 2010). This pattern suggests that objective performance indicators alone cannot account for observed exit behavior.

Moreover, variation in exit decisions may stem from differences in how entrepreneurs interpret information. Belief distortions may not only delay exit but also influence which exit modes are considered viable or desirable. Exit timing depends on how entrepreneurs assess continuation value and the potential for future learning under uncertainty (Dixit & Pindyck, 1994). If these assessments are systematically biased, exit thresholds become subjective and may shift upward. As a result, entrepreneurs may persist.

The next chapter therefore examines overconfidence as a central cognitive bias and analyzes how it shapes exit timing by influencing perceived continuation value under uncertainty.

3 Theoretical Background II: Overconfidence in the Context of Entrepreneurship

Building on the identified research gap, this chapter introduces overconfidence as a central cognitive bias in entrepreneurship. It shows how overconfidence shapes entrepreneurial decision-making, with particular attention to exit behavior under uncertainty.

3.1 Conceptual Clarification and Distinguishing Related Concepts

This section clarifies the concept of overconfidence and distinguishes it from related constructs. It further frames overconfidence as a belief-based bias that systematically affects expectation formation under uncertainty.

Overconfidence is a systematic bias in individuals' self-assessments. Overconfident individuals overestimate their abilities, knowledge, or the accuracy of their judgments. Notably, overconfidence is not a stable personality trait. Rather, it is a context-dependent cognitive bias that is especially likely in uncertain environments (Moore & Healy, 2008).

From a cognitive perspective, overconfidence is best understood as a situationally activated bias. While individual characteristics such as optimism or narcissism may correlate with overconfident beliefs, the degree of overconfidence varies systematically with information and task structure as well as environmental uncertainty (Moore & Healy, 2008).

High uncertainty and ambiguous feedback reduce the informativeness of outcomes, which makes belief miscalibration more likely to arise and persist (Soll et al., 2015). This is especially relevant for analyzing dynamic decisions such as persistence and exit (Dixit & Pindyck, 1994; Hayward et al., 2006).

Busenitz and Barney (1997) provide early evidence that entrepreneurs differ systematically from managers in large organizations in their reliance on heuristics and confidence judgments under uncertainty. Entrepreneurs rely more heavily on heuristic decision rules and exhibit stronger confidence in their assessments. While this heuristic orientation may facilitate opportunity recognition and rapid action in uncertain environments, it may also increase susceptibility to biased belief formation. Importantly, their analysis does not examine how such belief distortions affect dynamic belief updating or the timing of disinvestment decisions. The present thesis extends this line of inquiry by linking overconfidence to exit timing and exit mode choice within a dynamic decision framework.

Conceptually, overconfidence differs from optimism and self-efficacy. Optimism refers to a generalized tendency to expect positive outcomes, independent of a clearly specified causal pathway or objective probability assessment (Hmieleski & Baron, 2009). Self-efficacy refers to beliefs about one's ability to perform specific tasks or actions successfully (Bandura, 1997). Overconfidence is different from these concepts because it is defined by miscalibration between subjective beliefs and objective accuracy and can distort probability judgments and information interpretation (Moore & Healy, 2008; Ucbasaran et al., 2010).

Within entrepreneurial settings, overconfidence can influence occupational choice and subsequent decisions by distorting subjective expectations about success (Camerer & Lovallo, 1999). Individuals facing similar external conditions can form systematically different beliefs about their likelihood of success. Accordingly, this can explain market entry despite high average failure rates (Camerer & Lovallo, 1999; Koellinger et al., 2007).

In addition, beliefs that are distorted by overconfidence shape expectations about future cash flows and risk. Evidence from behavioral and corporate finance indicates that overconfident decision-makers systematically overestimate expected returns and underestimate downside risk (Bernardo & Welch, 2001; Malmendier & Tate, 2015). This can lead to distorted investment and disinvestment choices.

Moreover, it is important to distinguish overconfidence from confidence per se. High confidence can be warranted when it reflects superior skill or information. Overconfidence, by contrast, refers to systematic miscalibration between subjective beliefs and objective accuracy (Moore & Healy, 2008). This distinction is critical because the behavioral implications arise not from confidence itself but from persistent belief errors that distort decision-making under uncertainty.

3.2 Three Forms of Overconfidence

This section introduces the three forms of overconfidence: overestimation, overplacement, and overprecision. It outlines their conceptual differences and discusses their relevance for entrepreneurial decision-making.

Overconfidence is inherently multidimensional. Moore and Healy (2008) distinguish between three distinct forms of overconfidence, i.e., overestimation, overplacement, and overprecision. These forms capture different ways in which beliefs deviate from objective benchmarks. Each form reflects a specific type of miscalibration and can have different implications for entrepreneurial decision-making. Importantly, Moore and Healy (2008) emphasize that these

forms do not necessarily co-occur and can have distinct behavioral consequences depending on the decision context.

3.2.1 Overestimation

Overestimation refers to the tendency to overestimate one's actual ability, performance, level of control, or probability of success. Its core feature is absolute miscalibration. Subjective self-assessments exceed objective benchmarks and do not depend on comparisons with other people (Moore & Healy, 2008).

Notably, evidence of overestimation appears across domains. These include performance judgments (e.g., students overestimating exam results) and forecasting or diagnostic accuracy (e.g., physicians overestimating diagnostic precision), as well as control and planning judgments (Moore & Healy, 2008).

In entrepreneurial settings, overestimation has been linked both to market entry and to persistence under unfavorable conditions. Individuals who overestimate their probability of success are more likely to enter competitive markets with objectively high average failure rates and low average returns (Koellinger et al., 2007). This inflated self-assessment increases the perceived expected value of entrepreneurial activity and lowers subjective entry thresholds.

Importantly, the same mechanism can affect post-entry decisions (Hayward et al., 2006). For example, when entrepreneurs overestimate their chances of success or recovery, negative performance signals may be interpreted as temporary setbacks. Experimental evidence suggests that such belief distortions contribute to continued commitment, which implies persistence beyond economically optimal levels (Sandri et al., 2010). Taken together, overestimation influences the decision to enter and systematically shifts subjective continuation thresholds.

Entry decisions are particularly sensitive to overestimation because they require individuals to form expectations about venture prospects under substantial uncertainty. If these expectations are systematically inflated, entrepreneurship appears more attractive than it objectively is. Koellinger et al. (2007) argue that differences in subjective success probabilities play a central role in explaining entry decisions. This finding is consistent with experimental evidence showing excess entry driven by overconfidence (Camerer & Lovallo, 1999; Koellinger et al., 2007). This mechanism implies that distorted beliefs do not merely affect entry rates but also shape how entrepreneurs evaluate expected returns more generally. If overestimation increases the likelihood that individuals perceive entry as worthwhile, it may likewise raise subjective exit thresholds.

Inflated success probabilities also translate into biased valuations. By assigning excessively high success probabilities, overconfident individuals overestimate the expected value of entrepreneurial investments. This logic can account for continued investment of resources in projects with negative expected value when evaluated against objective benchmarks. This mechanism is highlighted in theoretical work and supported by empirical evidence linking entrepreneurial overconfidence to excess entry and low survival rates (Bernardo & Welch, 2001; Koellinger et al., 2007).

From a financial perspective, overconfidence affects expected returns as well as the perceived distribution of future cash flows. Overestimation inflates the subjective mean of expected outcomes, while overprecision generates excessive certainty about those expectations. This leads entrepreneurs to underestimate the range of potential downside realizations (Bernardo & Welch, 2001; Moore & Healy, 2008). As a result, investments may appear attractive even when risk-adjusted expected returns would be negative under more realistic assumptions. This distorted valuation logic is especially consequential in dynamic decisions such as entrepreneurial persistence and exit, where continuation preserves optionality but also prolongs exposure to losses (Dixit & Pindyck, 1994; Folta & O'Brien, 2004).

Further, overestimation is not necessarily static. In dynamic environments, miscalibration can persist or even intensify if individuals update beliefs asymmetrically. Entrepreneurs often overweight positive signals and interpret negative outcomes as temporary or externally driven (Moore & Healy, 2008). Because entrepreneurial performance signals are frequently noisy and ambiguous, such biased updating can sustain inflated success expectations over time. Overestimation may therefore continue to influence decision-making well beyond the initial entry stage, contributing to delayed exit and prolonged continuation (Hmieleski & Baron, 2009; Sandri et al., 2010).

In addition, empirical studies rarely isolate the different forms of overconfidence. For example, Koellinger et al. (2007) provide indirect evidence of absolute overconfidence, showing that high (self-assessed) entrepreneurial ability and strong expectations of success coexist with high aggregate average failure rates and low base rates of success. Although they do not explicitly distinguish between absolute and relative overconfidence, this pattern suggests systematic miscalibration. Their findings align with experimental evidence by Camerer and Lovallo (1999), who document excess entry in competitive settings driven by overly optimistic relative performance beliefs. Taken together, these studies indicate that entrepreneurial confidence frequently combines inflated success expectations with overly favorable self-assessment relative to others.

3.2.2 Overplacement

Overplacement is the tendency to overestimate one's position relative to others. This phenomenon is commonly described by the "better-than-average" effect. Unlike overestimation, overplacement is relational and rooted in social comparison (Moore & Healy, 2008; Svenson, 1981).

A common pattern of overplacement is that a majority of people rate themselves above the median on desirable attributes. This outcome is mathematically impossible. A well-known example is a study by Svenson (1981). Participants in the study rated their driving ability compared to others. Most believed they were more skilled and drove less risky than the median driver of their country. In entrepreneurship, overplacement arises when founders view themselves as more capable than competing entrepreneurs. They may also underestimate the strength of rivals (Moore & Healy, 2008).

Overplacement is especially important in competitive environments where outcomes depend on relative performance (Camerer & Lovallo, 1999). Individuals may enter markets not because expected average returns are high but because they believe they will outperform competitors facing the same conditions. Relative overconfidence therefore contributes to excessive entry. This effect is demonstrated experimentally and formalized in competitive settings, and consistent with field evidence on entrepreneurial entry and survival patterns (Camerer & Lovallo, 1999; Koellinger et al., 2007).

Further, the implications extend beyond entry. Relative overconfidence can lead entrepreneurs to pursue more aggressive strategies and systematic underestimation of competitors (Camerer & Lovallo, 1999; Moore & Healy, 2008). Entrepreneurs who perceive their ventures as superior to comparable firms may overestimate standalone value and undervalue external exit options. As a result, founders may reject or delay acquisition or sale opportunities that would appear attractive under more realistic comparisons (Cumming & MacIntosh, 2003; Hayward et al., 2006). This mechanism is consistent with why exit may be postponed even when viable and value-maximizing alternatives are available (DeTienne et al., 2015; Gimeno et al., 1997).

Moreover, ambiguity in reference groups reinforces overplacement. Entrepreneurs are rarely required to specify precise benchmarks. This flexibility allows selective comparison with weaker or non-comparable peers and generates systematic upward distortions in relative self-assessment, even without deliberate bias (Camerer & Lovallo, 1999; Moore & Healy, 2008).

3.2.3 Overprecision

Overprecision refers to excessive certainty about the accuracy of one's beliefs. It is reflected in confidence intervals that are too narrow and in systematic underestimation of uncertainty (Moore & Healy, 2008).

Overprecision is common in numerical forecasting under uncertainty. Individuals frequently display excessive certainty when forecasting outcomes despite limited or noisy information. In entrepreneurial contexts, this may manifest as strong confidence in estimates of demand or market size. Such excessive certainty produces rigid expectations that are difficult to revise over time (Hmieleski & Baron, 2009; Moore & Healy, 2008).

A key consequence of overprecision is slow belief updating. Overprecision affects not the arrival of information but the weight assigned to new signals in belief revision (Keren & Wu, 2015; Moore & Healy, 2008). New signals gradually reduce uncertainty and lead decision-makers to update their beliefs as evidence accumulates. When entrepreneurs perceive their forecasts to be highly precise, contradictory information is more likely to be treated as temporary noise (Hmieleski & Baron, 2009). Belief updating therefore becomes asymmetric. Favorable signals are readily incorporated, while unfavorable signals are discounted or rationalized (Soll et al., 2015). In entrepreneurial environments, such belief rigidity can sustain optimistic expectations despite deteriorating fundamentals and persistent negative feedback, thereby increasing the likelihood of prolonged commitment and persistence beyond economically optimal thresholds (Hayward et al., 2006; Hmieleski & Baron, 2009; Sandri et al., 2010).

Notably, overprecision concerns the perceived reliability of beliefs instead of their level. Individuals may therefore hold overly precise beliefs even when average expectations are not strongly biased (Moore & Healy, 2008). In entrepreneurial contexts, excessive certainty can lead to rigid expectations about market development or timing of exit. This rigidity reduces the likelihood that new information may contribute to substantial belief revision, thereby amplifying the effects of other forms of overconfidence in dynamic decision-making (Hmieleski & Baron, 2009; Moore & Healy, 2008).

Although overestimation, overplacement, and overprecision are conceptually distinct, they may interact in practice. Overestimation can inflate beliefs about absolute success probabilities, while overplacement reinforces the belief that one will outperform competitors facing similar conditions. Overprecision can further strengthen these effects by increasing certainty about the accuracy of such beliefs and reducing responsiveness to disconfirming information. In combination, these

biases can generate particularly persistent belief distortions over time. Recognizing these interactions is important for understanding why overconfidence may become especially consequential in dynamic entrepreneurial decision-making processes (Hayward et al., 2006; Moore & Healy, 2008).

Overall, overestimation, overplacement, and overprecision capture complementary dimensions of overconfidence. While they often co-occur in practice, they are conceptually and empirically separable. A multidimensional view of overconfidence is therefore essential for understanding how confidence shapes entrepreneurial decision-making (Moore & Healy, 2008).

3.3 Overconfidence in Entrepreneurship Research

This section reviews empirical and theoretical research on overconfidence in entrepreneurship. It synthesizes evidence on how biased beliefs affect entry, persistence, escalation of commitment, and venture performance.

Empirical work across surveys and experiments suggests that overconfidence matters throughout the entire entrepreneurial process. It does not influence a single decision. Overconfidence shapes how entrepreneurs evaluate opportunities and interpret feedback (Busenitz & Barney, 1997; Moore & Healy, 2008). Experimental approaches have been advocated as a valuable lens for studying entrepreneurial decision-making under uncertainty, as they allow researchers to isolate cognitive biases and behavioral frictions from purely economic incentives (Camerer & Lovallo, 1999).

Entrepreneurial decision-making under uncertainty is characterized by greater reliance on heuristics and biased judgments, which shapes how beliefs and expectations are formed (Busenitz & Barney, 1997). Risk preferences play a minor role in this context. Entrepreneurial environments often provide ambiguous and delayed feedback, which makes subjective expectations particularly influential. Overconfidence can therefore shape initial choices as well as belief updating and the decision-making process as a whole (Busenitz & Barney, 1997; Moore & Healy, 2008).

The effects of overconfidence vary across individuals and contexts. Factors such as industry uncertainty or prior entrepreneurial experience may moderate the strength and persistence of biased beliefs. While experience can improve calibration in some settings, prior success may also reinforce overconfident beliefs, particularly with respect to perceived ability or forecasting certainty (Hayward et al., 2006; Moore & Healy, 2008).

3.3.1 Overconfidence and Entry Decisions

A robust finding is that overconfidence increases the likelihood of entrepreneurial entry. Starting a venture involves high uncertainty. Under such conditions, individuals who overestimate their abilities or venture prospects are more likely to perceive entrepreneurship as attractive (Camerer & Lovallo, 1999; Koellinger et al., 2007).

Moreover, studies show that entrepreneurs tend to hold overconfident beliefs about their probability of success. These beliefs may exceed objective benchmarks and, in competitive settings, exceed the expected performance of other potential entrants. For example, overestimation inflates perceived expected returns, whereas overplacement leads individuals to believe they are more capable than competing entrepreneurs. Together, these biases can increase entry rates, including entry into highly competitive markets with low average success probabilities (Camerer & Lovallo, 1999; Koellinger et al., 2007; Moore & Healy, 2008).

Additionally, Koellinger et al. (2007) emphasize that entrepreneurial entry is driven less by differences in objective risk exposure and more by differences in subjective success expectations. While their analysis focuses on ex-ante success expectations and entry decisions, it falls short of directly addressing belief updating subsequent to entry. This limitation motivates extending the analysis to dynamic post-entry decisions such as persistence and exit. Existing work suggests that belief distortions shaping entry decisions may persist beyond entry and influence subsequent continuation choices (Camerer & Lovallo, 1999; Sandri et al., 2010). Individuals can face similar external conditions and still make different entry decisions because they assign different probabilities to success. This implies that overconfidence affects entry mainly through distorted beliefs rather than superior information or greater risk tolerance.

A related implication concerns base-rate neglect. Overconfident individuals may disregard average failure rates and instead focus on their perceived personal ability or the supposed uniqueness of their venture. This bias helps explain why entrepreneurial entry remains prevalent. The pattern has been demonstrated experimentally and is consistent with field (Camerer & Lovallo, 1999; Koellinger et al., 2007).

Further, overconfidence at the entry stage may serve as an instrumental function. By reducing perceived risk and increasing subjective success probabilities, it can motivate action under uncertainty. In some settings, fully unbiased assessments might discourage entry. Overconfidence can therefore help explain why entrepreneurial activity persists despite high average failure rates (Bernardo & Welch, 2001; Camerer & Lovallo, 1999).

3.3.2 Overconfidence, Persistence, and Escalation of Commitment

While Chapter 2 documented the existence of exit inertia and escalation behavior, this section explains how overconfidence provides a belief-based mechanism that sustains such persistence.

Overconfidence plays a key role in entrepreneurial persistence. After founding a venture, entrepreneurs must repeatedly interpret new information and decide whether to continue with the operations or exit. Overconfidence affects this process because it biases how feedback is perceived and integrated into expectations about future performance (Busenitz & Barney, 1997; Moore & Healy, 2008).

Moreover, overestimation can lead entrepreneurs to explain poor outcomes as temporary or related to external factors instead of core problems. Overplacement can reinforce beliefs that the venture will outperform competitors despite evidence to the contrary. Overprecision can increase certainty in forecasts and turnaround strategies. This can slow belief updating when new information becomes available (Camerer & Lovallo, 1999; Moore & Healy, 2008).

These effects are stronger because entrepreneurial feedback is often noisy and delayed (Hmieleski & Baron, 2009). Entrepreneurs may believe that poor performance reflects unfavorable market conditions. This makes it harder for entrepreneurs to disentangle bad luck from bad judgment. Under such conditions, overconfident beliefs may persist and remain uncorrected, contributing to delayed responsiveness to negative information and prolonged continuation (Moore & Healy, 2008; Sandri et al., 2010).

These persistence dynamics are reinforced by biased belief updating. While entrepreneurs are repeatedly exposed to new performance signals, overprecision affects how these signals are interpreted instead of whether they are observed. Entrepreneurs who are overly confident in the precision of their judgments tend to underweight negative feedback and attribute poor outcomes to temporary shocks or external conditions (Moore & Healy, 2008; Soll et al., 2015). Because entrepreneurial environments rarely provide clear or timely signals, this selective updating can sustain commitment even as cumulative evidence increasingly points toward underperformance (Busenitz & Barney, 1997; Hayward et al., 2006). Persistence therefore reflects not only optimism about future outcomes but also rigidity in belief revision that slows learning and increases the risk of escalation over time (Gimeno et al., 1997; Staw, 1981).

Further, entrepreneurial learning is complicated by asymmetric feedback. Signals of success are often more salient and easier to interpret than signals of failure, which may be attributed to temporary shocks or unfavorable conditions. This asymmetry creates scope for overconfident

belief maintenance, as negative information can be discounted or reinterpreted. Under such conditions, biased beliefs may persist, which reinforces persistence and escalation of commitment (Moore & Healy, 2008; Shepherd, 2003).

From a financial perspective, persistence can be interpreted as a capital allocation decision. Continued investment in a venture requires entrepreneurs to reassess expected returns relative to alternative uses of capital and effort. Overconfidence may distort this reassessment by sustaining overly optimistic expectations, which leads to continued investment in projects with declining or negative expected value (Bernardo & Welch, 2001; Malmendier & Tate, 2015). Although Malmendier and Tate (2015) study corporate managers, the underlying belief-based mechanisms are directly applicable to entrepreneurial decision-making, where feedback is noisier and governance constraints are weaker.

These dynamics connect to the concept of escalation of commitment (Staw, 1981). Decision-makers may continue to invest in failing courses of action because of prior investments and biased beliefs. In entrepreneurial contexts, overconfidence can increase the likelihood that founders remain committed beyond economically optimal exit points (Gimeno et al., 1997; Shepherd, 2003).

3.3.3 Overconfidence and Venture Performance

The relationship between overconfidence and performance is complex. Overconfidence can support success in early stages. It may increase motivation and resilience of the entrepreneur or the willingness to pursue bold strategies. These qualities can be useful in uncertain and dynamic environments (Bernardo & Welch, 2001; Hmieleski & Baron, 2009).

At the same time, the same bias can become harmful over time. As more performance information becomes available, overconfidence may prevent learning and delay necessary adjustments. Overestimation can lead to overly aggressive investments. Overplacement can lead to underestimating competitors. Overprecision can reduce responsiveness to negative feedback (Hmieleski & Baron, 2009). Empirical work suggests that moderate confidence can be beneficial, while excessive overconfidence is linked to resource misallocation and worse long-term outcomes (Malmendier & Tate, 2015).

Overall, the effects of overconfidence depend on context and timing. Overconfidence can be beneficial when it encourages action and experimentation. It can hurt when it prevents accurate learning and timely adjustment (Hmieleski & Baron, 2009; Moore & Healy, 2008).

From a financial perspective, the performance effects of overconfidence should be evaluated in risk-adjusted terms. Overconfident entrepreneurs may pursue strategies with high upside potential as well as substantial downside risk. While such strategies can generate exceptional outcomes in some cases, they may also increase performance volatility without improving expected returns. Consequently, observed performance differences may reflect greater risk exposure and return volatility rather than superior risk-adjusted performance or value creation (Bernardo & Welch, 2001). This pattern mirrors evidence from corporate finance. Malmendier and Tate (2015) show that overconfident managers increase investment intensity and performance volatility without necessarily improving risk-adjusted returns.

Importantly, exit decisions differ significantly from entry. Exit requires entrepreneurs to revise prior beliefs. It may also require accepting that earlier expectations were wrong (Hayward et al., 2006; Moore & Healy, 2008; Shepherd, 2003). These psychological costs imply that overconfidence may play a strong role in delaying exit or shaping the chosen exit mode.

Building on the real options logic discussed in Chapter 2, exit decisions resemble real option problems (Dixit & Pindyck, 1994). Continuing a venture preserves the option to recover future value but it also requires ongoing investment and exposure to downside risk. Overconfidence may inflate perceived continuation value and reduce sensitivity to downside outcomes, which leads entrepreneurs to delay exit beyond economically optimal points (Dixit & Pindyck, 1994; Sandri et al., 2010).

Literature consistently documents the presence of overconfidence across entrepreneurial contexts. It increases market entry, shapes persistence, and has mixed effects on venture performance. However, the role of overconfidence in entrepreneurial exit decisions remains less developed, and limited attention has been paid to how biased beliefs affect disengagement from a venture (DeTienne, 2010; Wennberg & DeTienne, 2014). Exit decisions are dynamic and made under uncertainty (Dixit & Pindyck, 1994). The decision-not-to-decide framework by Schade and Snir (2020) which will be introduced in Chapter 4 provides the decision-theoretic structure within which these belief distortions translate into systematic exit postponement.

These features of exit decisions make them a particularly relevant setting for overconfidence. This gap motivates the following analysis, which links overconfidence to exit timing and exit modes. Prior research suggests that exit decisions are particularly sensitive to cognitive framing and belief distortions, as they require entrepreneurs to reassess future value under uncertainty (Moore & Healy, 2008).

Taken together, these findings suggest that overconfidence has systematic financial consequences beyond entrepreneurial entry. By distorting expectations, overconfidence alters the perceived value of continuation relative to exit. Biased beliefs can inflate the subjective option value of waiting while compressing perceived downside risk (Bernardo & Welch, 2001; Malmendier & Tate, 2015). This perspective provides a direct conceptual bridge between the entrepreneurship literature on overconfidence and real options models of exit timing.

Conceptually, these belief distortions shift the subjective continuation threshold (Dixit & Pindyck, 1994; Folta & O'Brien, 2004). Overestimation raises perceived expected cash flows, overprecision compresses perceived downside risk, and overplacement inflates perceived standalone value relative to market-based exit alternatives. In a real options view, these distortions increase the perceived option value of waiting and delay the moment at which exit becomes acceptable (Dixit & Pindyck, 1994).

3.4 Operationalizing Overconfidence in Entrepreneurship

This section reviews common approaches to measuring overconfidence in empirical research. It discusses measurement challenges and outlines how different forms of overconfidence can be operationalized in entrepreneurial contexts.

Empirical research uses several approaches to measure overconfidence. The variety of different approaches reflects the complexity of the construct. Because overconfidence appears as a systematic difference between perceived beliefs and reality, measures often compare subjective assessments with objective benchmarks. Different forms of overconfidence require different measurement logics. No single measure captures all dimensions (Keren & Wu, 2015; Moore & Healy, 2008).

Measuring overconfidence poses challenges because observed responses may reflect both genuine belief distortions and measurement error. Self-reported assessments may be noisy or influenced by strategic responding or ambiguity in survey questions. As a result, empirical studies often rely on indirect or behavioral measures to infer overconfidence. Interpreting such measures requires caution, as they capture systematic patterns at the aggregate level rather than precise belief distortions at the individual level (Keren & Wu, 2015; Soll et al., 2015).

Moreover, overestimation is commonly measured by comparing individuals' subjective perceptions with subsequently observed outcomes or externally verifiable indicators. In Koellinger et al. (2007), respondents report their perceived entrepreneurial skill and expectations about opportunities. These subjective expectations are not evaluated against realized outcomes at the

individual level. Instead, the authors compare the prevalence of optimistic perceptions across countries with aggregate entrepreneurial outcomes observed ex post, e.g., start-up rates and proxy measures of business survival. Systematic gaps between expectations and outcomes are consistent with the presence of biased beliefs that affect exit timing and exit mode (Koellinger et al., 2007).

In entrepreneurial finance, these measures map to outcomes such as cash flows, venture survival, or realized returns. Exit events constitute the primary mechanism through which financial returns are realized, particularly in venture capital-backed firms (Cumming & MacIntosh, 2003). They are therefore central to understanding investment efficiency and exit timing decisions.

Further, overplacement is typically measured with relative self-assessments as in the study of Svenson (1981). Individuals are asked to rank their abilities or expected performance relative to others in the reference group. If most respondents place themselves above the median, this pattern suggests relative overconfidence. This approach captures the social-comparative nature of overplacement and fits in competitive contexts such as entrepreneurship (Moore & Healy, 2008; Svenson, 1981).

In addition, overprecision is often measured using confidence intervals around numerical estimates or forecasts. Individuals are considered to be overprecise when their confidence intervals are systematically too narrow and therefore fail to contain realized outcomes as frequently as implied by their stated confidence levels (Moore & Healy, 2008). This measurement approach captures excessive certainty about beliefs rather than biased belief levels. In entrepreneurial contexts, overprecision manifests as unwarranted confidence in forecasts (Hmieleski & Baron, 2009; Soll et al., 2015).

Additionally, empirical studies often find that the three measures of overconfidence are only weakly correlated. This suggests that overconfidence is not a single latent trait. It is better understood as a set of related but separable biases (Moore & Healy, 2008). The choice of measurement should therefore depend on the specific decision context and the theoretical mechanism of interest.

Further, measuring overconfidence in the context of entrepreneurial exit poses challenges. Exit decisions are forward-looking and depend on expectations about venture prospects (DeTienne, 2010). Overconfidence may therefore manifest not only in inflated performance expectations but also in excessive certainty regarding turnaround prospects and in biased comparisons with potential exit alternatives (Moore & Healy, 2008). Recognizing these measurement implications

is essential for developing a coherent theoretical framework linking overconfidence to entrepreneurial exit decisions.

In the context of entrepreneurial exit, overconfidence may manifest in several observable ways. These include delayed exit relative to industry benchmarks, continued investment despite declining performance indicators, rejection of acquisition offers that exceed comparable valuations, or excessive certainty regarding future recovery prospects (Gimeno et al., 1997; Malmendier & Tate, 2015). Such outcomes can be interpreted as indirect evidence of biased beliefs affecting exit timing and exit mode.

Moreover, the three forms of overconfidence can affect entrepreneurial exit decisions through distinct but complementary belief-based mechanisms (Hayward et al., 2006; Moore & Healy, 2008). For example, overestimation inflates perceived probabilities of future success and raises the perceived continuation value of the venture. Overplacement distorts relative evaluations by reducing the perceived attractiveness of external exit options such as sales or acquisitions (e.g., M&A) compared to continued independence. Overprecision increases certainty about forecasts and turnaround strategies, which slows belief updating in response to negative information (Moore & Healy, 2008). Together, these mechanisms shift subjective exit thresholds upward and make continued operation and postponement more attractive. This mapping clarifies how overconfidence translates into delayed exit in dynamic decision environments, consistent with experimental and empirical evidence on disinvestment inertia (Gimeno et al., 1997; Sandri et al., 2010).

While the literature documents how overconfidence shapes beliefs and persistence, it remains unclear how such biased beliefs translate into structured exit behavior under uncertainty (D. DeTienne & Wennberg, 2016; Wennberg & DeTienne, 2014). The following chapter introduces the decision-not-to-decide framework by Schade and Snir (2020) to provide the decision-theoretic structure within which overconfidence operates.

4 Theoretical Bridge: Schade & Snir (2020) and the Decision-Not-to-Decide

This chapter introduces the decision-not-to-decide framework by Schade and Snir (2020) and uses it to bridge economic and behavioral perspectives on entrepreneurial exit under uncertainty.

4.1 The Decision-Not-to-Decide under Uncertainty

This section presents the decision-not-to-decide framework and contrasts it with standard interpretations of exit delay based on status quo bias and escalation of commitment.

Classical decision theory assumes that decision-makers choose the option that maximizes expected utility based on their beliefs and preferences. When circumstances change, rational agents are expected to update their beliefs and actions accordingly. From this benchmark perspective, postponing an exit decision from an underperforming venture, once expected continuation value falls below available alternatives, would be interpreted as inefficient or as evidence of biased behavior (Dixit & Pindyck, 1994; McDonald & Siegel, 1986).

A common explanation for such delay is status quo bias. Samuelson and Zeckhauser (1988) show that individuals decide disproportionately often for the status quo alternative. Individuals tend to prefer the current state over change, even in scenarios where alternative options offer higher expected utility. Persistence may reflect risk aversion, loss aversion, regret avoidance, or the presence of sunk costs. In entrepreneurship research, continued commitment to weak ventures has therefore often been explained as inertia or escalation of commitment instead of as the outcome of strategic reasoning (D. R. DeTienne et al., 2008; Staw, 1981).

Schade and Snir (2020) challenge this interpretation by rethinking the meaning of inaction. Instead of treating delayed exit as a symptom of status quo bias or simple inertia, they conceptualize postponement as a behavioral decision outcome that reflects how individuals cope with irreversible commitment under uncertainty. They term this phenomenon the “decision not to decide”.

Their decision-not-to-decide framework implies that delay can arise even without explicit optimization or deliberate strategic planning. Postponement is not merely passive inertia but a systematic behavioral response to decisional stress and commitment under uncertainty (Schade & Snir, 2020).

Importantly, deferred commitment does not necessarily imply payoff-maximizing behavior. Schade and Snir (2020) do not claim that postponement is generally optimal or strategically planned. Instead, they show evidence suggesting that delay emerges systematically from how

individuals cope with decisional stress and irreversibility. Observed behavior can diverge from real options predictions, despite the option value of waiting. From a real options perspective, postponement is more defensible when waiting is expected to generate informative signals (Dixit & Pindyck, 1994; McDonald & Siegel, 1986). In Schade and Snir's (2020) framework, however, postponement can also emerge as a coping-related decision outcome.

From a financial perspective, postponement differs substantially from immediate exit in its implications for expected cash flows and the timing of value realization. Immediate exit corresponds to exercising a disinvestment option and locking in current value. Delaying exit preserves flexibility by maintaining the option to abandon later, while remaining exposed to future upside and downside risk (Dixit & Pindyck, 1994; McDonald & Siegel, 1986).

4.1.1 Decision Modes and the Timing of Commitment

In their experimental design, Schade and Snir (2020) operationalize the decision-not-to-decide framework through two linked choices: an initial investment or participation decision (whether to invest and enter the game at all) and, conditional on participation, a repeated disinvestment timing decision (when to exit). Their empirical analysis focuses on how decision-makers differ in these choices relative to a real options benchmark.

Building on this structure, exit behavior can be interpreted as reflecting different decision modes. Each mode reflects a different stance toward the timing and irreversibility of commitment. For analytical clarity, this thesis distinguishes two modes. These modes are not intended as an exhaustive typology but as an analytically convenient way to capture differences in preferences. The distinction between decision modes is introduced here as an analytical abstraction to interpret observed behavior. It is not a formal element of Schade and Snir's (2020) experimental model.

In this thesis, "exit mode" refers to the route of exit introduced by DeTienne et al. (2015), e.g., financial harvest exits, whereas "decision mode" refers to the timing of irreversible commitment (immediate exit vs. postponement).

Decision mode A: Immediate irreversible commitment to exit (non-entry or immediate disinvestment)

In the experimental setting, this decision mode corresponds either to skipping the game or disinvesting at the earliest possible round after entry. This phase is intended to simulate the first step of a real life investment decision. In this scenario, control is relinquished and potential future upside is foregone. Uncertainty is resolved immediately but flexibility is lost. From a financial

perspective, this can be interpreted as keeping the investment sum and potentially reallocating resources elsewhere.

Decision mode B: Postponement of irreversible commitment

After entering, the entrepreneur continues operating the venture and defers irreversible exit. Thereby, flexibility is preserved, including the option to abandon at a later point. From a real options perspective, such postponement can reflect anticipated learning or the hope that future signals will clarify whether continuation is worthwhile (Dixit & Pindyck, 1994; McDonald & Siegel, 1986).

By contrast, Schade and Snir (2020) emphasize that postponement does not need to be driven by expected learning or strategic optimization. In their framework, delay can arise as a systematic behavioral response to decisional stress under uncertainty, reflecting how individuals cope with the burden of committing to an irreversible course of action. Financially, continued operation retains the option value of waiting while deferring the realization of losses or gains.

Within this postponement mode, motives may differ. In some cases, delay is driven by the option value of waiting for additional information under uncertainty (Dixit & Pindyck, 1994; McDonald & Siegel, 1986). In other cases, postponement reflects a preference to avoid the psychological and identity-related costs associated with irreversible commitment, e.g., loss of role identity or anticipated grief reactions, rather than a belief that waiting will necessarily generate superior information (Shepherd, 2003). Both variants are treated here as instances of postponed irreversible commitment. They differ in their underlying motivation, not the nature of the decision itself (DeTienne & Wennberg, 2016; Wennberg & DeTienne, 2014)

This two-mode abstraction mirrors the central contrast emphasized by Schade and Snir (2020). Exit decisions are not merely choices between action and inaction but choices between committing now and deferring commitment under uncertainty.

The decision-not-to-decide framework of Schade and Snir (2020) is conceptually different from escalation-of-commitment explanations. Escalation theories emphasize backward-looking forces, e.g., sunk costs and prior investments that psychologically bind decision-makers to failing courses of action (Staw, 1981). By contrast, the decision not to decide is forward-looking (Schade & Snir, 2020). This distinction clarifies why postponement can occur even in the absence of large sunk costs and why continued engagement does not need to imply irrational attachment to past decisions.

Although forward-looking postponement is conceptually distinct from escalation of commitment, repeated postponement can increase sunk investments over time and potentially give rise to escalation-like persistence (Gimeno et al., 1997; Staw, 1981).

A central implication of combining real options logic with the decision-not-to-decide framework of Schade and Snir (2020) is that postponement can be rational under certain conditions. When uncertainty is high, commitments are irreversible and future learning is expected, delaying decisions preserves flexibility. This logic is well established in the real options literature, which shows that the option to wait has positive value under precisely these conditions (Dixit & Pindyck, 1994; McDonald & Siegel, 1986).

At the same time, postponement in their framework does not reflect indecision or disengagement. Entrepreneurs remain actively involved in the venture while deferring final commitment. Thus, delay is best understood as a choice over timing, not as an absence of choice (Schade & Snir, 2020).

In real options logic, postponement is particularly attractive when uncertainty is high, decisions are irreversible, and additional information is expected to arrive (Dixit & Pindyck, 1994; McDonald & Siegel, 1986). Schade and Snir (2020) add that postponement may still occur systematically as a coping-related decision outcome under decisional stress. From a real options perspective, committing too early can lock decision-makers into outcomes that later prove suboptimal. Postponement preserves the possibility of benefiting from favorable future developments while avoiding premature commitment (Dixit & Pindyck, 1994).

Beyond economic considerations, related work highlights psychological drivers of postponement. Decisions involving responsibility and potential regret are especially likely to be delayed (Shepherd, 2003). Exiting a venture often entails not only financial consequences but also psychological, social, and reputational costs. When such commitment costs are salient, entrepreneurs may prefer to defer irreversible decisions even if expected outcomes do not improve with waiting (Schade & Snir, 2020). Postponement reflects a temporal preference for delaying costly commitment rather than a belief that waiting will necessarily generate superior information.

The decision-not-to-decide framework of Schade and Snir (2020) does not imply that exit postponement is always efficient. While real options theory justifies delay when uncertainty is high and additional learning is expected, postponement may become inefficient when belief updating is distorted and subjective continuation values remain inflated. Accordingly,

distinguishing between option-based flexibility and belief-driven persistence is central to understanding entrepreneurial exit timing.

Moreover, the scope for postponement is constrained by external factors. Liquidity constraints, debt obligations, investor pressure, or large competitive and technological shocks can force entrepreneurial exit even in situations where entrepreneurs prefer to delay commitment. Such shocks may sharply reduce the informational value of waiting by resolving uncertainty abruptly, thereby undermining the rationale for postponement (Cefis et al., 2022; Cumming & MacIntosh, 2003).

The framework developed by Schade and Snir (2020) is particularly relevant for entrepreneurship because exit is a process. Entrepreneurs typically face repeated opportunities to reconsider exit while operating under uncertainty and incomplete feedback (DeTienne, 2010; Wennberg & DeTienne, 2014). The ability to postpone commitment while remaining engaged closely mirrors the realities of entrepreneurial decision-making. By explicitly modeling decision modes rather than single decisions, the framework of Schade and Snir (2020) captures the temporal and processual nature of exit in a way that static models cannot.

Importantly, Schade and Snir (2020) do not argue that biased beliefs are the primary driver of postponement. Their contribution lies in demonstrating that delay can arise as a systematic behavioral outcome that may, under some conditions, be consistent with rational timing under uncertainty. The present chapter builds on this structure by examining how biased belief formation, when present, may interact with these decision modes and influence exit timing and exit mode choice.

4.1.2 Empirical Grounding: Experimental Evidence from Schade & Snir (2020)

The decision-not-to-decide framework developed by Schade and Snir (2020) is grounded in controlled experimental evidence. To examine whether postponement can be understood as a systematic behavioral outcome, the authors conduct a laboratory experiment with real monetary incentives that closely mirrors a real options disinvestment problem.

In their laboratory experiment, Schade and Snir (2020) study repeated investment and disinvestment decisions under uncertainty. At the outset, participants decide whether to enter the decision situation at all (i.e., whether to invest or skip participation). Conditional on participation, they repeatedly choose whether to continue operating the investment project or to disinvest and realize the current value. The economic environment is designed such that, under standard real options logic, a well-defined optimal stopping rule exists. This rule specifies when exit should

occur given the evolution of outcomes. This allows the authors to classify observed behavior as closer to or farther from optimal disinvestment timing.

A central feature of the experimental design is the measurement of individual decision-making styles. Drawing on Janis and Mann's (1977) conflict theory of decision-making, Schade and Snir (2020) distinguish between vigilance and buck-passing, as stable ways of coping with decisional stress. Vigilant individuals are able to tolerate the stress associated with difficult decisions. They process available information carefully and tend to commit once sufficient evidence indicates that a decision should be taken. Moreover, vigilance is associated with timely decision-making and closer adherence to normative benchmarks.

By contrast, buck-passing reflects a maladaptive coping style in which decision-makers experience decision-making as stressful and seek to avoid it (Schade & Snir, 2020). Buck-passers prefer to delegate or postpone decisions instead of committing to an irreversible course of action. In cases where delegation to others is not possible, buck-passers tend to shift decisions into the future, effectively postponing commitment (Janis & Mann, 1977). In the context of Schade and Snir's (2020) experiment, this tendency manifests as systematic exit delay relative to the real options benchmark. These decision styles are measured independently of the experimental task and treated as stable individual characteristics.

The experimental results show systematic heterogeneity in exit behavior that cannot be fully explained by standard real options models alone. Participants with higher buck-passing tendencies are significantly more likely to postpone disinvestment and to continue operating the investment project beyond the point at which exit would be optimal according to the real options benchmark. In contrast, more vigilant participants tend to exit closer to the optimal threshold. These differences are economically meaningful. Delayed exit among buck-passers increases the likelihood of lower average payoffs relative to participants whose behavior aligns more closely with the benchmark (Schade & Snir, 2020).

The authors further show that these behavioral tendencies already matter at the entry stage. Individuals who are prone to buck-passing are more likely to avoid participation altogether, consistent with the idea that decision-makers may anticipate future difficulties in making timely exit decisions. This pattern is compatible with planner-doer interpretations of intertemporal decision-making (Thaler & Shefrin, 1981). Risk aversion moderates these effects, indicating that preferences over risk interact with decision styles in shaping both entry and exit behavior (Schade & Snir, 2020).

Taken together, these findings provide direct empirical support for the decision-not-to-decide framework (Schade & Snir, 2020). They demonstrate that exit postponement is not random noise. Instead, it is systematically related to individual differences in how decision-makers cope with uncertainty and decisional stress. This empirical grounding is crucial for the present thesis. It establishes postponement as a meaningful and observable decision mode (Decision mode B in the present framework) and motivates the subsequent analysis of how belief formation and cognitive biases may further influence the perceived value of waiting and the timing of entrepreneurial exit.

4.2 Behavioral Real Options Interpretation of Exit Postponement

This section interprets the decision-not-to-decide framework of Schade and Snir (2020) through a behavioral real options perspective and links postponement to disinvestment logic under uncertainty.

The decision-not-to-decide framework of Schade and Snir (2020) can be understood as a behavioral complement to real options theory. Both approaches emphasize the value of flexibility under uncertainty and the costs of irreversible commitment (Dixit & Pindyck, 1994). However, while real options theory typically focuses on optimal stopping rules given fixed beliefs, Schade and Snir (2020) explicitly allow for heterogeneity in how decision-makers approach commitment and exit timing. Rather than assuming that exit decisions are governed by a mechanically applied threshold, their framework highlights that individuals differ in how they cope with uncertainty and the stress associated with committing to a final decision. This makes their framework particularly well suited to integration with behavioral real options approaches that recognize the role of subjective beliefs and cognitive processes in shaping perceived option value.

As discussed in Chapter 2, entrepreneurial exit is a dynamic disinvestment decision. Continuing a venture preserves the possibility of benefiting from favorable future developments, whereas exit irreversibly eliminates this option and fixes outcomes at the time of disinvestment. The value of waiting depends on expected payoffs, volatility, and the costs of delay. Exit becomes optimal once the expected value of continuation falls below a critical threshold (Dixit & Pindyck, 1994; McDonald & Siegel, 1986).

Moreover, standard real options models often characterize this decision using a single boundary. Above the threshold, continuation dominates, whereas below it, immediate exit is optimal. These models assume rational expectations and unbiased belief updating as new information arrives. Within this framework, postponement is optimal only to the extent that it maximizes expected value given objective conditions (Dixit & Pindyck, 1994).

A behavioral interpretation suggests a more nuanced decision structure. Rather than assuming that entrepreneurs adhere closely to a sharp threshold, observed behavior may exhibit systematic postponement relative to the real options benchmark (Dixit & Pindyck, 1994). Although Schade and Snir (2020) do not formally model an inaction region, their framework is compatible with the idea that delaying irreversible commitment can arise systematically when uncertainty remains high and the psychological and economic costs of exit are salient.

Empirical evidence reviewed in Chapter 2 shows that entrepreneurs frequently delay exit (Sandri et al., 2010). The decision-not-to-decide framework helps explain this pattern by suggesting that postponement itself may be systematically observed when uncertainty persists and commitment is costly (Schade & Snir, 2020). In this interpretation, the value of waiting depends not only on objective conditions but also on how decision-makers perceive uncertainty, learning potential, and future upside (Moore & Healy, 2008). This extends Schade and Snir's (2020) framework by allowing subjective beliefs to shape the perceived option value of continuation.

A key distinction in this context is between objective and subjective learning. Objective learning refers to the arrival of new information that genuinely reduces uncertainty about future outcomes. Subjective learning concerns how decision-makers interpret and integrate such information into their beliefs (Moore & Healy, 2008). Even when objective signals are weak or uninformative, entrepreneurs may perceive waiting as valuable because they expect future clarity or improvement (Hmieleski & Baron, 2009). Consequently, postponement can remain attractive.

From a behavioral real options perspective, the perceived option value of waiting is shaped by both economic incentives and psychological interpretation. Entrepreneurs form expectations about venture prospects. If these expectations are biased, the perceived value of continuation may be inflated. In such cases exit is delayed because the option to wait is perceived as especially valuable (Moore & Healy, 2008).

Evidence from corporate finance supports this interpretation. Malmendier and Tate (2015) show that overconfident managers systematically overestimate future cash flows and underestimate downside risk. This leads them to invest more aggressively and to persist longer in underperforming projects. Applied to entrepreneurship, similar belief distortions can inflate the perceived option value of waiting and delay exit decisions beyond economically optimal thresholds (Hayward et al., 2006; Sandri et al., 2010).

The perceived option value of waiting is shaped by entrepreneurs' own belief formation processes. Entrepreneurs' expectations influence how attractive postponement appears. If these beliefs are

overly optimistic, postponement can become self-reinforcing. Continued engagement is justified by optimistic expectations and the absence of exit is interpreted as preserving valuable flexibility (Moore & Healy, 2008). In this way, behavioral distortions can transform the option to wait from a contingent opportunity into a dominant strategy.

Disinvestment decisions are particularly susceptible to such distortions because exit is irreversible and can be closely tied to identity and responsibility. Leaving a venture may involve relinquishing control and abandoning potential future upside. These features heighten concerns about regret and personal accountability, making postponement psychologically relevant (Shepherd, 2003). In this context, delaying exit allows entrepreneurs to defer responsibility for negative outcomes while preserving the possibility of future success (Schade & Snir, 2020). While Schade and Snir (2020) emphasize decisional stress and coping with irreversible commitment, entrepreneurship research highlights psychological and identity-related costs of exit, e.g., grief (Shepherd, 2003). Postponement can consequently serve as a response to both decisional stress and broader psychological costs of exit.

This perspective reframes exit postponement as a choice among temporal strategies. Entrepreneurs may justify waiting with reference to learning or market recovery (Moore & Healy, 2008). Postponement avoids premature commitment while maintaining a sense of control over future outcomes, consistent with the decision-not-to-decide framework (Schade & Snir, 2020).

Moreover, a central implication is that exit timing depends on subjective thresholds shaped by belief formation. Classical disinvestment models predict exit once continuation value crosses a given threshold (Dixit & Pindyck, 1994). A behavioral interpretation suggests that belief formation and learning processes influence where this threshold is perceived to be (Moore & Healy, 2008). As a result, entrepreneurs facing similar venture trajectories may choose different timings of exit depending on how they perceive uncertainty and the value of waiting (Dixit & Pindyck, 1994; Gimeno et al., 1997).

While the decision-not-to-decide framework of Schade and Snir (2020) focuses on individual-level decision processes under uncertainty, exit timing is also shaped by external constraints. Liquidity limitations, investor pressure, debt covenants, or contractual obligations can restrict the feasibility of prolonged postponement and force exit (Cefis et al., 2022; Cumming & MacIntosh, 2003). Such constraints impose financial boundaries on the decision not to decide, even though the framework itself abstracts from detailed institutional settings (Schade & Snir, 2020).

Additionally, exit timing may also exhibit path dependence. Repeated postponement decisions can alter the subsequent decision environment by increasing sunk investments, reinforcing prior beliefs, and narrowing the range of perceived alternatives (Gimeno et al., 1997; Staw, 1981). While this dynamic resembles escalation of commitment, the key distinction is that persistence here arises from forward-looking preferences over deferring irreversible commitment under uncertainty. Delaying exit today can therefore make exit less likely tomorrow.

In summary, interpreting exit postponement through a behavioral real options lens bridges economic and psychological explanations of entrepreneurial exit. Exit delay can be interpreted as arising from the desire to preserve option value and from biased belief formation at the same time (Moore & Healy, 2008). This provides a foundation for analyzing how specific cognitive biases, e.g., overconfidence, may inflate the perceived value of waiting. The next section builds on this interpretation by examining the role of overconfidence in shaping exit timing and exit mode choice.

4.3 Integrating Overconfidence into the Decision-Not-to-Decide Framework

This section integrates overconfidence into the decision-not-to-decide framework of Schade and Snir (2020). It argues that biased belief formation systematically increases the attractiveness of exit postponement. While Schade and Snir's (2020) framework specifies the structure of exit decisions under uncertainty, the present section extends their framework by introducing overconfidence as a cognitive mechanism that shapes how entrepreneurs evaluate and select among available decision modes.

Schade and Snir (2020) conceptualize exit decisions as choices over the timing of irreversible commitment. They distinguish between immediate exit and postponement of irreversible commitment. Their framework does not treat biased beliefs as a core assumption or starting point. Instead, it demonstrates that postponement can arise as a systematic behavioral pattern under uncertainty, without relying on systematic belief distortions (Schade & Snir, 2020). Building on this structure, the present analysis introduces overconfidence as a belief-based mechanism that operates within the same decision framework and helps explain why postponement may become persistently attractive.

Overconfidence distorts beliefs about venture prospects, learning potential, relative performance, and the controllability of outcomes. By biasing expectations, overconfidence inflates the perceived value of waiting and raises the subjective threshold at which exit becomes acceptable (Moore &

Healy, 2008). In entrepreneurship research, overconfidence is typically understood as a systematic distortion of beliefs and not as a simple preference (Hayward et al., 2006). This makes it particularly relevant for analyzing exit decisions (DeTienne, 2010).

Several alternative explanations for exit postponement have been proposed, including risk preferences and loss aversion, status quo bias, and escalation of commitment to prior investments (Samuelson & Zeckhauser, 1988; Shefrin & Statman, 1985; Staw, 1981). While these mechanisms capture important aspects of persistence behavior, they primarily emphasize stable preferences or backward-looking attachment rather than the formation and updating of forward-looking expectations. Overconfidence is uniquely suited to fill this gap because it operates directly on belief formation (Hayward et al., 2006; Moore & Healy, 2008). By distorting expectations, overconfidence shapes how entrepreneurs evaluate the option value of waiting and how attractive postponement appears relative to exit.

Further, evidence from corporate finance supports this interpretation. Overconfident managers tend to delay disinvestment and persist in underperforming projects, which is consistent with inflated continuation values (Malmendier & Tate, 2015). Applied to entrepreneurship, similar belief distortions can inflate the perceived option value of waiting and delay exit beyond economically optimal thresholds (Hayward et al., 2006).

In addition, overconfidence may also intensify over time. Continued survival, even in the absence of strong performance improvements, can be interpreted as confirmation of underlying ability or strategy. Prior investments and effort of the entrepreneur further increase commitment and confidence in eventual success (Gimeno et al., 1997; Staw, 1981). Postponement itself may reinforce overconfident beliefs, creating a feedback loop in which delay sustains optimism and optimism sustains delay (Moore & Healy, 2008).

Importantly, overconfidence does not replace the decision-not-to-decide framework of Schade and Snir (2020). It operates within it by shaping how decision-makers evaluate the attractiveness of postponement relative to the decision of exit.

4.3.1 Overestimation and Exit Postponement

Overestimation leads entrepreneurs to overstate their probability of future success and their ability to influence outcomes (Moore & Healy, 2008). It may also lead them to exaggerate the effectiveness of potential turnaround strategies. Within a behavioral real options perspective, these inflated expectations increase the perceived upside of waiting and raise the subjective threshold at which exit becomes acceptable (Dixit & Pindyck, 1994). Even when objective indicators point to

declining venture prospects, entrepreneurs may believe that success remains likely if they persist long enough (Hayward et al., 2006).

Within the decision-not-to-decide framework of Schade and Snir (2020), overestimation increases the subjective attractiveness of postponement. Exit delay can arise as a systematic behavioral outcome under uncertainty, even when additional learning is unlikely to be profitable and immediate exit would be optimal from a real options perspective. When entrepreneurs overestimate future success or recovery prospects, postponement can be framed as giving the venture "more time". Overestimation therefore reinforces the preference for delaying irreversible commitment within the existing decision structure rather than constituting an independent explanation of exit delay (Moore & Healy, 2008).

4.3.2 Overplacement and Exit Mode Choice

Overplacement affects exit decisions by biasing entrepreneurs' relative evaluations (Moore & Healy, 2008). Entrepreneurs who believe they are superior to other founders or that their venture performance is above average may systematically undervalue exit alternatives, e.g., sale or liquidation. This affects exit mode choice (DeTienne et al., 2015). Within the decision-not-to-decide framework of Schade and Snir (2020), overplacement makes postponement attractive because exit options are perceived as relatively inferior to continued independence.

Additionally, exit offers may be rejected or deferred because entrepreneurs believe that better opportunities will emerge later or that current valuations underestimate the venture's true potential (Hayward et al., 2006). Importantly, this mechanism does not require entrepreneurs to deny weak absolute performance. It operates through relative overconfidence. Even financially struggling entrepreneurs may believe that competitors are worse off or that they possess unique capabilities that justify continued engagement (Moore & Healy, 2008).

4.3.3 Overprecision and Exit Inertia

Overprecision influences exit decisions by generating excessive certainty about venture prospects (Moore & Healy, 2008). Entrepreneurs may underestimate uncertainty and express unwarranted confidence in specific forecasts, e.g., when market conditions will improve or when strategic pivots will succeed (Hmieleski & Baron, 2009).

Within a behavioral real options framework, overprecision slows belief updating. Negative performance signals or external feedback may be dismissed as temporary noise and not as informative evidence (Moore & Healy, 2008). Entrepreneurs may repeatedly postpone exit while remaining confident in earlier expectations. This produces behavioral exit inertia. Rather than fully

reassessing continuation, decision-makers remain anchored in prior beliefs and defer irreversible commitment over time.

4.3.4 Synthesis

Taken together, the three forms of overconfidence reinforce the decision not to decide. Overestimation inflates expected future returns, overplacement distorts comparisons between continuation and exit alternatives, and overprecision slows belief updating in the face of adverse information (Moore & Healy, 2008). Each mechanism increases the perceived option value of waiting and raises the subjective threshold at which exit becomes acceptable (Dixit & Pindyck, 1994; Moore & Healy, 2008).

Further, the interaction of these biases further strengthens exit postponement. Overestimation sustains optimistic expectations, overplacement weakens the perceived attractiveness of exit options, and overprecision reduces responsiveness to negative signals (Moore & Healy, 2008). Together, these mechanisms help explain why exit postponement can persist even when any single bias alone would not fully account for prolonged delay.

These mechanisms fit naturally into the decision structure proposed by Schade and Snir (2020). Overconfidence does not alter the set of available decision modes but it systematically influences how entrepreneurs evaluate them. By biasing beliefs, overconfidence makes exit postponement increasingly attractive relative to exit (Hayward et al., 2006; Moore & Healy, 2008). This synthesis sets the stage for the final step of the theoretical argument. If overconfidence systematically increases exit postponement through distinct cognitive mechanisms, then clear propositions linking different forms of overconfidence to entrepreneurial exit timing and exit mode choice can be derived. The next chapter presents these propositions explicitly.

5 Propositions

Building on the literature of entrepreneurial exit, overconfidence, and the decision-not-to-decide framework (Schade & Snir, 2020), this chapter develops a set of propositions that formalize how biased belief formation shapes entrepreneurial exit behavior under uncertainty. The focus is on how overconfidence affects both exit timing and exit mode choice when entrepreneurs face uncertainty, irreversible decisions, and evolving expectations.

The propositions integrate three core theoretical streams. First, the exit literature highlights that exit is a dynamic and forward-looking decision with multiple possible outcomes (DeTienne, 2010; DeTienne et al., 2015). Second, research on overconfidence shows that entrepreneurs form systematically biased beliefs regarding (a) the level of venture prospects (overestimation), (b) relative standing compared to competitors (overplacement), and (c) the precision of their forecasts (overprecision) (Moore & Healy, 2008). Third, the decision-not-to-decide framework emphasizes that postponement can be a choice under uncertainty (Schade & Snir, 2020). Taken together and consistent with Chapters 2, 3, and 4, these perspectives imply that exit behavior depends not only on objective venture performance or real options benchmarks but also on how entrepreneurs interpret information and evaluate continuation compared to exit alternatives.

The dominant case considered in the propositions is exit delay. In line with real options logic, delaying exit can be optimal (Dixit & Pindyck, 1994). The following propositions argue that overconfidence can inflate the perceived value of waiting and shift subjective exit thresholds. Consequently, entrepreneurs may postpone exit beyond economically optimal points. This mechanism represents the central focus of the following theoretical argument.

Importantly, earlier exit is not ruled out. It is treated as a boundary case. In particular, relative overconfidence and optimistic beliefs about outside options may, under certain conditions, increase the likelihood of early exit. These cases are acknowledged but are not the primary focus of the present thesis.

Several factors are intentionally not modeled. This analysis does not explicitly incorporate cross-regional or cultural differences in exit behavior. It also abstracts from general risk preferences and institutional variation except where governance directly constrains discretion over exit timing (boundary condition in P6). These factors may influence exit decisions but the focus here is on belief formation and belief updating, not on stable preferences or contextual heterogeneity. By holding these elements constant, the propositions isolate the role of overconfidence as a cognitive mechanism.

The propositions presented hereafter provide testable implications linking distinct forms of overconfidence to observable exit outcomes. Together, they offer a coherent framework for analyzing how biased beliefs affect entrepreneurial disinvestment decisions under uncertainty. The following propositions distinguish mechanisms that primarily affect exit timing (P1, P3, P4) from those that primarily affect exit mode choice (P2, P5), noting that mode-choice mechanisms also have timing implications through the acceptance or postponement of exit offers. Finally, P6 and P7 introduce systematic heterogeneity in these effects.

5.1 Core Mechanisms: Overconfidence and Exit Delay

Prior research in entrepreneurship has consistently shown that entry decisions are strongly shaped by subjective success expectations (Camerer & Lovallo, 1999; Koellinger et al., 2007). Koellinger et al. (2007) demonstrate that individuals differ systematically in their perceived probability of entrepreneurial success. These subjective beliefs explain entry decisions even when objective average failure rates are high. Their findings suggest that entrepreneurship is best understood as a belief-based choice under uncertainty.

Entrepreneurial market entry often requires entrepreneurs to assign sufficiently high subjective success probabilities to justify action under uncertainty, sometimes in ways that depart from objective base rates (Camerer & Lovallo, 1999; Koellinger et al., 2007). The present thesis argues that this belief-based logic does not end with entry. Moreover, it extends the logic from the entry stage to the exit stage of the entrepreneurial process. Once a venture is operating, entrepreneurs repeatedly reassess whether continued investment is justified or whether exit represents the value-maximizing alternative. These reassessments are inherently forward-looking and depend on expectations about venture prospects (DeTienne, 2010; Dixit & Pindyck, 1994; Gimeno et al., 1997). If beliefs are biased at entry, these beliefs can also bias disinvestment decisions.

Proposition 1

Overestimation reflects absolute miscalibration of beliefs. Entrepreneurs systematically overestimate their prospects of success, e.g., future recovery or their ability to influence outcomes. While overestimation may indirectly affect how information is interpreted, its primary mechanism operates through inflated expected return levels (Moore & Healy, 2008).

In real option terms, exit becomes optimal when the expected continuation value falls below the value of exit (Dixit & Pindyck, 1994). Overestimation directly affects this mechanism. By inflating perceived expected cash flows and exaggerating recovery prospects, it raises the subjective continuation value of the venture relative to objective performance signals. As a result, the

subjective exit threshold shifts upward compared to an unbiased benchmark (Dixit & Pindyck, 1994; Hayward et al., 2006).

In addition, this mechanism is conceptually different from escalation of commitment. Escalation emphasizes backward-looking attachment to prior investments and sunk costs (Staw, 1981). Overestimation, by contrast, is forward-looking. It distorts expectations about future returns rather than reinforcing psychological attachment to past expenditures (Moore & Healy, 2008). The entrepreneur does not persist primarily because of sunk costs but because anticipated future outcomes are systematically overstated.

Further, the distinction becomes particularly salient in ventures exhibiting continuous underperformance, e.g., sustained negative cash flows or declining revenues. From a standard disinvestment perspective, such signals should gradually reduce expected continuation value and trigger exit once alternative opportunities dominate (Dixit & Pindyck, 1994). However, entrepreneurs who overestimate their likelihood of recovery may interpret negative signals as noisy or externally caused. They may believe that success remains likely if they persist long enough and "hope for sunshine" (Moore & Healy, 2008).

Within the decision-not-to-decide framework (Schade & Snir, 2020), this distortion increases the relative attractiveness of postponing irreversible commitment (Decision mode B) over immediate exit (Decision mode A). Overestimation does not alter the available decision modes. Instead, it systematically biases how they are evaluated by maintaining elevated subjective continuation values (Moore & Healy, 2008). Consequently, even when objective performance indicators suggest that further waiting provides limited incremental value, entrepreneurs may perceive continued operation as justified.

Importantly, this mechanism does not imply that delayed exit is inevitable. If negative information becomes sufficiently precise and salient, belief miscalibration may eventually be corrected (Moore & Healy, 2008). Nevertheless, as long as inflated expectations persist, the dominant behavioral implication of overestimation is exit delay under sustained underperformance.

Taken together, extending this belief-based logic from the entry to the exit stage leads to the following proposition:

P1: Entrepreneurs who overestimate their venture's likelihood of future recovery or success are more likely to delay exit from ventures exhibiting persistent underperformance, relative to an unbiased real options benchmark.

Proposition 2

This proposition concerns exit mode choice rather than exit timing per se, although postponement of financially attractive exits necessarily implies delayed exit.

Entrepreneurial decision-making often unfolds in competitive environments in which outcomes depend on relative performance (Camerer & Lovallo, 1999; Moore & Healy, 2008). Individuals frequently overestimate their position relative to others. In entrepreneurial contexts, overplacement arises when founders believe their ventures are superior to competing firms or undervalued by the market (Koellinger et al., 2007; Moore & Healy, 2008).

Unlike overestimation, which inflates the level of expected returns, overplacement distorts relative comparisons (Moore & Healy, 2008). It affects how entrepreneurs position their venture against competitors and market benchmarks.

Moreover, exit decisions often require comparison between alternatives, e.g., continuing independently, selling the venture, or liquidating assets (DeTienne et al., 2015; Wennberg et al., 2010). These alternatives are inherently comparative. Entrepreneurs must assess whether the subjective value of continued independence exceeds the value offered by external exit options, e.g., sale prices. While market prices may themselves be imperfect, they constitute salient external benchmarks reflecting relative valuation assessments (DeTienne et al., 2015).

Importantly, overplacement systematically distorts this comparison. Entrepreneurs who believe their ventures are above average are more likely to perceive market-based exit offers as undervaluing their venture's true potential (Koellinger et al., 2007; Moore & Healy, 2008). By inflating perceived relative standing, overplacement increases the subjective value of continued independence relative to externally offered prices. Rejection of such offers therefore reflects distorted comparative evaluation.

Although overplacement could theoretically encourage early exit if it extends primarily to outside opportunities, e.g., overconfidence in launching a superior new venture or securing attractive employment (DeTienne, 2010; Headd, 2003), this represents a boundary case. Such an effect requires that relative overconfidence applies more strongly to alternative opportunities than to their current venture, which is unlikely. Accordingly, early exit driven by overplacement should be understood as an asymmetric and less frequent outcome.

Within the decision-not-to-decide framework (Schade & Snir, 2020), overplacement increases the relative attractiveness of continued independence and postponement (Decision mode B) compared to immediate exit, e.g., through sales or IPOs (Decision mode A). It does so by biasing comparative

valuation rather than by inflating expected cash flows directly. The dominant behavioral implication is therefore rejection or deferral of externally attractive exit opportunities.

These arguments lead to the following proposition:

P2: Entrepreneurs who overplace their ventures relative to competitors are less likely to accept externally offered financial exit options that are objectively attractive based on performance benchmarks and are more likely to continue operating independently, thereby postponing exit.

Proposition 3

Entrepreneurial exit decisions require continuous belief updating as new performance information becomes available (DeTienne, 2010). In standard real options models, learning gradually reduces uncertainty and adjusts the expected continuation value of a venture. Exit becomes optimal once accumulated information lowers the expected value of continuation below a critical threshold (Dixit & Pindyck, 1994).

Chapter 3 conceptualized overprecision as a distinct form of overconfidence which is characterized by excessive certainty regarding the accuracy of one's beliefs. Unlike overestimation, which inflates the level of expected returns, overprecision distorts the confidence attached to those expectations. It narrows subjective confidence intervals and reduces the weight assigned to new information relative to prior beliefs (Moore & Healy, 2008).

Exit inertia refers to repeated postponement of exit decisions despite the accumulation of negative performance signals (Sandri et al., 2010). Overprecision provides a cognitive mechanism for such inertia. Because entrepreneurs exhibit unwarranted confidence in the precision of their forecasts, e.g., expectations of recovery or turnaround success, they are more likely to interpret adverse signals as temporary noise and not as evidence that requires substantial belief revision (Hmieleski & Baron, 2009; Moore & Healy, 2008).

Although direct evidence linking overprecision to entrepreneurial exit is limited, related findings support the proposed mechanism. Overconfident managers respond sluggishly to adverse information and persist longer in underperforming projects (Malmendier & Tate, 2015). Experimental evidence documents disinvestment inertia even when additional waiting yields limited informational value (Sandri et al., 2010). These findings are consistent with excessive certainty weakening responsiveness to negative signals.

Applied to entrepreneurship, overprecision interferes with the learning process that underlies optimal exit timing. By assigning insufficient weight to new adverse information, entrepreneurs

slow the decline in subjective continuation value. As a result, the exit threshold implied by unbiased updating is reached later than economically optimal. Resources remain committed beyond the point at which expected NPV would turn negative under realistic assumptions (Dixit & Pindyck, 1994).

Within the decision-not-to-decide framework (Schade & Snir, 2020), overprecision increases the likelihood of repeated postponement (Decision mode B) by delaying the shift toward exit (Decision mode A). While excessive certainty could theoretically produce premature exit when pessimistic beliefs are held with equal rigidity, empirical evidence indicates that optimism and overconfidence are positively correlated in entrepreneurial contexts (Cooper et al., 1988; Ucbasaran et al., 2010). Accordingly, the dominant behavioral implication of overprecision is delayed exit through slowed belief updating.

These arguments lead to the following proposition:

P3: Entrepreneurs who exhibit overprecision are more likely to display exit inertia, characterized by repeated postponement or rejection of exit as negative performance signals accumulate.

5.2 Disinvestment Logic and the Decision-not-to-Decide

Schade and Snir (2020) conceptualize the decision not to decide as a mode of postponing irreversible commitment under uncertainty, which is driven by decisional stress and coping strategies, particularly by buck-passing and vigilance. The present thesis extends this framework by introducing overconfidence as a belief-based mechanism that systematically alters how decision modes are evaluated. Whereas Propositions 1-3 identify specific cognitive distortions in expected returns, relative comparisons, and belief updating, the present proposition integrates these distortions within a broader decision-mode framework. It suggests that overconfidence shifts entrepreneurs toward postponement as a preferred decisional coping strategy.

Proposition 4

Entrepreneurial exit decisions exhibit precisely the conditions under which postponement becomes attractive. As discussed in Chapter 2, exit is typically irreversible and undertaken under uncertainty (Dixit & Pindyck, 1994). Exit decisions require forward-looking evaluations of uncertain future returns (DeTienne, 2010) and often involve identity concerns and the potential realization of losses (Gimeno et al., 1997; Shepherd, 2003). These features increase both the economic and psychological costs of commitment.

From a behavioral real options perspective, overconfidence inflates the subjective value of flexibility. Entrepreneurs who are overconfident tend to overestimate future cash flows, underestimate downside risk, and exaggerate the likelihood of favorable developments for the venture and the market (Hayward et al., 2006; Moore & Healy, 2008). As a result, the continuation option appears especially valuable from their perspective. Postponing exit preserves this perceived option value while deferring irreversible commitment (Dixit & Pindyck, 1994). In decision-mode terms, overconfidence shifts the relative attractiveness of postponement (Decision mode B) over immediate exit (Decision mode A), independent of whether objective learning potential and incremental option value are limited (Sandri et al., 2010).

Importantly, this does not imply that waiting is inherently irrational. Real options logic recognizes the value of flexibility under uncertainty (Dixit & Pindyck, 1994). The distortion arises when subjective expectations systematically exaggerate the benefits of continued waiting relative to objective performance signals. In such cases, postponement persists beyond what would be justified by unbiased evaluation of informational gains.

In addition, overconfidence may increase the psychological costs of commitment. Exiting a venture requires acknowledging that earlier expectations may have been incorrect and overly optimistic, which can threaten the self-image or the identity of the entrepreneur (Hayward et al., 2006; Shepherd, 2003). Delaying exit allows entrepreneurs to defer this recognition while maintaining a sense of control over future outcomes. Thus, the decision not to decide serves both an economic function (preserving flexibility) and a psychological function (deferring irreversible acknowledgment of loss) (Schade & Snir, 2020).

Although overconfidence could, in principle, accelerate exit when pessimistic beliefs are held with excessive certainty, this represents a boundary condition (Cooper et al., 1988; Ucbasaran et al., 2010). This suggests that the dominant manifestation of overconfidence in exit decisions is delayed commitment.

These arguments lead to the following proposition:

P4: Overconfidence increases the likelihood that entrepreneurs adopt a decision-not-to-decide mode (Decision mode B), favoring postponement over immediate irreversible exit (Decision mode A), even when objective performance indicators imply that immediate exit would be optimal under real options benchmarks.

Proposition 5

The present proposition reframes entrepreneurial exit as an intertemporal capital allocation decision that can be systematically biased by overconfident continuation valuations.

At its core, entrepreneurial exit involves allocating scarce resources across alternative uses. Entrepreneurs must repeatedly evaluate whether continued investment yields higher expected value than exiting and reallocating resources to alternative opportunities (DeTienne, 2010; Dixit & Pindyck, 1994). Under standard financial reasoning, the exit mode that maximizes expected value, given available information and accounting for irreversibility and uncertainty, should be chosen. This includes accepting acquisition offers or pursuing IPOs when market-based valuations exceed the unbiased expected continuation value of the venture (Cumming & MacIntosh, 2003; DeTienne et al., 2015; Dixit & Pindyck, 1994).

Overconfidence, particularly overestimation and overplacement, systematically distorts this valuation comparison. By overstating future cash flows, exaggerating recovery potential, underestimating downside risk, or inflating relative standing (Hayward et al., 2006; Moore & Healy, 2008), entrepreneurs inflate the subjective continuation value of the venture relative to available harvest exit alternatives. As a result, financially attractive harvest exits may be perceived as undervaluing the venture's true potential. This can hold even when market-based offers exceed liquidation value or unbiased expected continuation value.

Additionally, evidence from corporate finance supports this mechanism. Overconfident managers persist longer in underperforming projects and respond sluggishly to adverse information (Malmendier & Tate, 2015). Behavioral finance research further documents reluctance to realize losses when subjective valuations exceed market benchmarks (Shefrin & Statman, 1985). Although these studies examine corporate settings, they illustrate how overconfident valuation can distort capital allocation decisions.

In entrepreneurial contexts, where feedback is noisier and governance constraints are looser, such distortions may be particularly consequential. When confronted with financially attractive harvest exits, overconfident entrepreneurs may systematically favor continued independence over capital redeployment. This behavior may reflect distorted intertemporal valuation (Gimeno et al., 1997). Resources remain committed to ventures whose continuation value is subjectively inflated relative to available exit alternatives.

Importantly, this mechanism concerns exit mode choice, not merely exit timing. Overconfidence does not only delay exit. It shifts preferences away from market-based harvest modes (e.g., sales or IPOs) toward continued independent operation.

This leads to the following proposition:

P5: Overconfident entrepreneurs are more likely to allocate capital toward continued independent operation rather than financial harvest exit modes, even when harvest exits are objectively financially attractive relative to performance-based benchmarks.

5.3 Individual Heterogeneity and Moderating Effects

The translation of overconfidence into exit delay is unlikely to be uniform across entrepreneurs. The strength and persistence of belief distortions depend on how individuals form, calibrate, and revise expectations under uncertainty. Accordingly, individual-level characteristics that systematically shape confidence calibration and responsiveness to feedback are expected to moderate the relationship between overconfidence and exit behavior (Wennberg et al., 2010).

While the core framework abstracts from demographic heterogeneity, gender differences and lifecycle characteristics provide theoretically grounded illustrations of how belief-based mechanisms may vary systematically across individuals. These moderators do not determine exit behavior directly but they shape the intensity with which overconfidence translates into postponement of irreversible commitment.

Proposition 6

Gender is introduced here not as a causal determinant of exit behavior but as a theoretically grounded proxy for systematic differences in confidence calibration and belief updating under uncertainty. A substantial body of research in behavioral economics and entrepreneurship shows that men and women differ, on average, in the level and calibration of confidence judgments. Men tend to report higher subjective success probabilities and exhibit stronger overestimation and overplacement. On the other hand, women's confidence judgments are, on average, more conservative and more closely aligned with objective benchmarks (Koellinger et al., 2013; Moore & Healy, 2008; Niederle & Vesterlund, 2007).

Importantly, these differences reflect variation in belief formation and updating. They do not imply differences in underlying ability (Koellinger et al., 2013; Moore & Healy, 2008). Experimental and field evidence suggests that men not only display higher initial confidence but also exhibit greater persistence of overconfident beliefs in the presence of ambiguous or noisy feedback

(Niederle & Vesterlund, 2007). Such persistence is particularly relevant in entrepreneurial contexts, where performance signals are delayed, noisy, and open to interpretation (Hmieleski & Baron, 2009).

Within the present framework, overconfidence raises the subjective continuation value of the venture and increases the likelihood of postponing irreversible exit (Propositions 1-4). However, the strength with which overconfidence translates into exit delay depends on the degree to which inflated beliefs are resistant to downward revision under negative feedback (Moore & Healy, 2008; Soll et al., 2015). When confidence is both upwardly biased and rigid in the face of adverse information, the translation of overconfidence into exit delay should be amplified.

To the extent that male entrepreneurs exhibit, on average, higher levels of overestimation and greater persistence of inflated beliefs, the positive relationship between overconfidence and exit delay is expected to be stronger among men. Female entrepreneurs, who on average display more conservative success expectations and greater responsiveness to negative performance signals, may exhibit a weaker amplification of subjective continuation value under comparable objective conditions (Koellinger et al., 2013; Moore & Healy, 2008).

This argument does not imply uniform behavior within gender groups, nor does it suggest that gender independently determines exit timing. Gender moderates the strength of the relationship between overconfidence and exit delay by shaping confidence calibration and responsiveness to information.

Notably, this moderating effect is subject to boundary conditions. In environments characterized by strong external governance, e.g., venture capital oversight or staged financing, individual belief distortions are more likely to be constrained (Cumming & MacIntosh, 2003). Under such conditions, belief distortions are more likely to be disciplined by formal decision structures, and the moderating role of gender should attenuate.

These considerations lead to the following proposition:

P6: The positive relationship between overconfidence and exit delay is, on average, stronger for male entrepreneurs than for female entrepreneurs, reflecting documented gender differences in confidence calibration and responsiveness to negative feedback under uncertainty.

Proposition 7

Entrepreneurial exit requires repeated belief updating in response to new information, as exit involves forward-looking evaluations of uncertain future returns (DeTienne, 2010; Dixit &

Pindyck, 1994). Persistence in underperforming ventures therefore reflects objective performance as well as responsiveness of belief updating processes (Gimeno et al., 1997; Hayward et al., 2006).

As argued in Proposition 3, overprecision reduces openness to belief revision by creating excessive certainty regarding the accuracy of prior expectations (Moore & Healy, 2008). Rather than inflating expected returns per se, overprecision narrows perceived confidence intervals and increases resistance to contradictory information. Negative performance signals are therefore more likely to be discounted as temporary noise (Sandri et al., 2010).

The behavioral consequences of overprecision, however, are unlikely to be uniform across entrepreneurs. Individual characteristics may condition how strongly epistemic rigidity translates into behavioral persistence (Wennberg et al., 2010). Entrepreneur age provides a theoretically grounded moderator.

Older entrepreneurs typically possess longer professional histories, more accumulated firm-specific experience, and more consolidated self-concepts tied to their ventures (Gimeno et al., 1997; Shepherd, 2003). Over time, entrepreneurial roles may become more central to personal identity, increasing the psychological cost of revising strongly held expectations about venture prospects. Research on cognitive entrenchment suggests that accumulated experience can stabilize mental models and increase resistance to disconfirming information (Dane, 2010). Under such conditions, excessive belief precision may be more difficult to correct once performance deteriorates.

Importantly, this argument does not assume that older entrepreneurs are inherently more overprecise or less capable decision-makers. Experience can improve calibration in many contexts. Rather, the claim is conditional. When overprecision is present, age-related identity consolidation and cognitive entrenchment may amplify its behavioral consequences. In other words, age strengthens the translation of epistemic rigidity into exit inertia.

At the same time, age may introduce countervailing forces. Shorter remaining career horizons could increase the attractiveness of earlier exit. Empirical evidence on age and entrepreneurial exit is mixed (Wennberg & DeTienne, 2014). The present argument therefore concerns a contingent effect. Under conditions of pronounced overprecision, identity-based and cognitive entrenchment mechanisms are expected to dominate horizon effects and increase persistence in underperforming ventures.

From a financial perspective, exit requires reassessing expected future returns relative to alternative uses of resources (Dixit & Pindyck, 1994). When overprecision sustains unwarranted

certainty about recovery prospects and age-related factors heighten resistance to belief revision, negative signals are less likely to trigger timely exit decisions. The result is repeated postponement of irreversible exit despite deteriorating performance.

This leads to the following proposition:

P7: The positive relationship between overprecision and exit inertia is stronger for older entrepreneurs than for younger entrepreneurs, on average and under comparable governance conditions.

6 Discussion

The following sections discuss the theoretical, practical, and research implications of treating entrepreneurial exit as a belief-based disinvestment process.

6.1 Theoretical Implications

The first key theoretical implication concerns how exit delay is conceptualized. Chapter 2 showed that entrepreneurial exit is best understood as a dynamic, forward-looking disinvestment decision characterized by uncertainty, irreversibility, and the gradual arrival of information (DeTienne, 2010; Dixit & Pindyck, 1994). Real options theory provides a normative benchmark for exit timing, emphasizing the value of flexibility and the option to wait (Dixit & Pindyck, 1994). However, empirical evidence reviewed in Chapter 2 demonstrates that entrepreneurs often delay exit beyond what can be justified by option value alone (Sandri et al., 2010).

This thesis contributes by showing that such delay can be understood as neither purely rational nor mere inertia. By integrating the decision-not-to-decide framework of Schade and Snir (2020) (Chapter 4), this thesis reframes exit postponement as a set of decision modes, i.e., as a choice over the timing of irreversible commitment. Notably, postponement in this framework reflects a systematic preference for deferring irreversible commitment under decisional stress. This perspective bridges the gap between rational real options models and behavioral explanations of persistence. Exit delay emerges as a response to uncertainty, whose attractiveness is shaped by the decision-maker's subjective beliefs.

A second contribution of this thesis lies in extending the decision-not-to-decide framework by explicitly incorporating belief-based mechanisms. Schade and Snir (2020) demonstrate that postponement can arise as a behavioral response to decisional stress under uncertainty. Their framework abstracts from assumptions about belief accuracy and instead focuses on how decisional stress shapes decisions of the entrepreneur. Notably, overconfidence does not replace the decisional stress and commitment-avoidance mechanisms emphasized by Schade and Snir (2020) but amplifies them by inflating the perceived benefits of postponement within the same decision structure. While decisional stress explains a preference for deferring commitment, overconfidence explains why postponement may be perceived as economically rational from the entrepreneur's perspective.

Chapters 3 and 5 extended this framework by introducing overconfidence as a mechanism that operates within this decision structure. Overconfidence does not create new decision modes but alters how entrepreneurs evaluate existing options, particularly the perceived value of postponing

irreversible commitment. This extension preserves the integrity of Schade and Snir's (2020) framework while increasing its explanatory power. It clarifies why postponement may persist even when learning opportunities are weak and real options benchmarks suggest timely exit.

A third theoretical contribution of this thesis concerns the distinction between exit timing and exit mode choice (DeTienne et al., 2015). The exit literature has long recognized heterogeneity in exit outcomes (Wennberg & DeTienne, 2014) but timing and mode are often not analytically distinguished. This thesis contributes by explicitly distinguishing mechanisms that primarily affect when entrepreneurs exit from those that primarily affect how they exit.

The propositions developed in Chapter 5 show that different forms of overconfidence map onto different exit outcomes. Overestimation and overprecision primarily affect exit timing by inflating perceived continuation value and slowing belief updating. Overplacement primarily affects exit mode choice by distorting comparisons between continued independence and external exit options, e.g., sale. Importantly, distortions of exit mode choice also have timing implications, as rejecting or postponing attractive exit offers necessarily delays exit.

This distinction advances theory by showing that exit delay is not a homogeneous phenomenon. Different cognitive biases can lead to superficially similar persistence behavior through distinct mechanisms, with different implications for performance and capital allocation.

More broadly, this thesis contributes to the emerging literature on behavioral real options. Traditional real options models assume rational expectations and unbiased learning. Behavioral extensions have acknowledged deviations but often lack a clear decision-theoretic structure. By combining real options logic with the decision-not-to-decide framework of Schade and Snir (2020) and belief-based distortions, this thesis offers a coherent behavioral real options interpretation of entrepreneurial exit.

Exit delay is thus understood as the outcome of two interacting forces: (1) the structural value of flexibility under uncertainty, and (2) biased beliefs that inflate the perceived option value of waiting. This integration helps reconcile seemingly conflicting findings in the literature.

6.2 Practical Implications

For entrepreneurs, the findings highlight that exit delay can result from strategic considerations and systematic biases in belief formation. Overconfidence can make postponement appear economically justified even when objective indicators suggest otherwise. Recognizing this tendency matters because delayed exit can impose substantial opportunity costs by tying up financial and emotional resources as well as time.

The framework developed in this thesis suggests that founders should explicitly separate hope for improvement from evidence-based learning. Structured decision rules, external feedback, and pre-commitment mechanisms (e.g., predefined exit thresholds) may help counteract biased belief updating and reduce inefficient delay. Notably, the results do not imply that earlier exit is always preferable. Rather, they emphasize the need for deliberate reflection on whether postponement is driven by genuine uncertainty or by inflated expectations.

For investors, particularly venture capitalists, the findings underscore the importance of mitigating belief-driven persistence. Investors often rely on entrepreneurs' judgments when assessing continuation versus exit (Cumming & MacIntosh, 2003). If these judgments are systematically biased, capital may remain allocated to ventures whose continuation no longer maximizes expected value.

Moreover, the distinction between exit timing and exit mode is particularly relevant for investors (Cumming & MacIntosh, 2003). For example, overplacement may lead founders to reject acquisition offers that are financially attractive from an investor's perspective. Understanding this bias can help align incentives and inform the design of staged financing arrangements or exit rights that reduce the scope for inefficient postponement.

At the policy level, the findings suggest that delayed exit should not automatically be interpreted as failure or incompetence of the entrepreneur. Postponement can reflect rational flexibility or belief distortions. Policies aimed at improving entrepreneurial outcomes could therefore focus not only on encouraging entry but also on facilitating informed exit decisions.

In addition, entrepreneurship education and support programs could place greater emphasis on exit planning and entrepreneurial decision-making under uncertainty. Reducing the stigma associated with exit and failure may lower psychological exit costs and reduce the incentive to postpone irreversible decisions purely to avoid loss realization or identity threats.

6.3 Limitations and Future Research

This thesis has several limitations that point to promising avenues for future research.

First, the analysis is entirely theoretical. While it builds on robust experimental and empirical findings, the propositions developed in Chapter 5 require empirical testing. Future research could test these propositions using longitudinal data on entrepreneurial ventures. Such work could combine measures of belief formation with observed exit timing and exit modes.

Further, this thesis abstracts from institutional and cultural variation. Exit behavior may differ systematically across different settings. Future work could examine how institutional contexts interact with belief-based mechanisms and decision modes.

Additionally, overconfidence is treated as the primary cognitive bias. Other biases, e.g., ambiguity aversion or self-efficacy, may also shape exit decisions. Future research could explore how multiple biases interact within the decision-not-to-decide framework of Schade and Snir (2020).

Finally, the analysis treats overconfidence as relatively stable over the decision horizon. In reality, belief distortions may evolve endogenously as entrepreneurs receive feedback or experience success and failure. Studying dynamic beliefs and decision modes represents an important next step.

Overall, this thesis contributes to a more nuanced understanding of entrepreneurial exit by showing how biased beliefs interact with the structure of irreversible decision-making under uncertainty.

7 Conclusion

This thesis examined why entrepreneurs often delay the decision to exit their ventures. Building on the entrepreneurial exit literature, research on overconfidence, and the decision-not-to-decide framework (Schade & Snir, 2020), this thesis developed a behavioral real options perspective on exit postponement. The core argument is that exit delay cannot be fully explained by rational option value or backward-looking escalation alone. Instead, it emerges from the interaction of irreversible decision structures and overconfidence-driven belief distortions that systematically inflate the perceived value of waiting.

This concluding chapter summarizes the key insights and articulates the overall contribution of this thesis.

7.1 Summary of Key Insights

This thesis proceeded in four main steps.

First, Chapter 2 established entrepreneurial exit as a dynamic, forward-looking disinvestment decision made under uncertainty, irreversibility, and evolving information (DeTienne, 2010; Dixit & Pindyck, 1994). Exit was conceptualized as a continuous timing problem (Wennberg & DeTienne, 2014). Entrepreneurs repeatedly compare the expected continuation value of the venture with available exit alternatives (DeTienne et al., 2015; Dixit & Pindyck, 1994). Real options theory provides a normative benchmark for this process by highlighting the value of flexibility and the option to wait (Dixit & Pindyck, 1994). At the same time, empirical evidence shows that entrepreneurs frequently delay exit beyond what can be justified by option value alone, which suggests the presence of systematic behavioral frictions in exit decisions (Sandri et al., 2010).

Further, Chapter 3 introduced overconfidence as a central cognitive bias shaping entrepreneurial decision-making under uncertainty. Distinguishing between overconfidence in its distinct forms, i.e., overestimation, overplacement, and overprecision, clarified that overconfidence is not a single construct but a set of belief distortions with distinct behavioral implications (Moore & Healy, 2008). Prior research has documented the role of overconfidence in entry, persistence, and escalation of commitment (Camerer & Lovallo, 1999; Koellinger et al., 2007; Staw, 1981). The third chapter extended these insights to the exit context by emphasizing that exit decisions are inherently belief-driven and forward-looking and therefore particularly susceptible to biased expectation formation and distorted belief updating (DeTienne, 2010; Moore & Healy, 2008). By

shifting the analytical focus from entry to disinvestment, this thesis repositioned overconfidence as a mechanism that shapes ventures beyond their creation and market entry.

Additionally, Chapter 4 introduced the decision-not-to-decide framework by Schade and Snir (2020) as a theoretical bridge between economic and behavioral decision-making. The framework reframes exit postponement as a choice over the timing of irreversible commitment and not as mere inertia or indecision. Postponement can arise as a systematic behavioral response to uncertainty and decisional stress, even when beliefs are unbiased. Integrating this framework with real options logic clarified that exit delay may be structured and meaningful without necessarily being optimal (Schade & Snir, 2020). This conceptual move allowed exit postponement to be interpreted as an active temporal strategy and not as a passive failure to act.

Moreover, Chapter 5 integrated overconfidence into the decision-not-to-decide framework of Schade and Snir (2020) and derived a set of propositions (P1-P7). These propositions link specific belief distortions to exit timing and exit mode choice and highlight heterogeneity among entrepreneurs.

Across these chapters, a consistent picture emerges. Exit behavior arises from the interaction of decision structures that make postponement feasible and sometimes valuable coupled with the subjective beliefs of entrepreneurs. Exit timing and exit mode choice are shaped by objective venture performance and by how entrepreneurs perceive uncertainty, learning potential, relative performance, and control.

7.2 Final Contribution Statement

The central contribution of this thesis is to offer an integrated explanation of entrepreneurial exit postponement that combines the decision-not-to-decide framework of Schade and Snir (2020) and belief-based biases. By doing so, it advances the entrepreneurship and behavioral decision-making literature.

This thesis contributes to entrepreneurship research. It reconceptualizes exit delay as a belief-based timing choice and not as a residual outcome or behavioral anomaly. It clarifies why entrepreneurs may rationally value flexibility while simultaneously delaying exit potentially inefficiently due to biased beliefs. This perspective helps reconcile conflicting interpretations of persistence as either rational waiting or escalation of commitment.

Additionally, this thesis extends the decision-not-to-decide framework (Schade & Snir, 2020) by explicitly incorporating belief formation. While Schade and Snir (2020) demonstrate that exit postponement can be a systematic behavioral response under uncertainty, this thesis shows how

overconfidence operates within the same decision structure to make postponement persistently attractive. Overconfidence does not alter the set of available decision modes. Instead, it systematically biases how entrepreneurs evaluate them.

Further, this thesis advances real options theory by showing that the option value of waiting is not purely economically constructed. It also has a psychological component. Subjective beliefs of decision-makers shape perceived continuation thresholds and exit decisions. This insight provides a foundation for future empirical research and links cognitive biases to dynamic investment and disinvestment behavior.

Taken together, this thesis demonstrates that entrepreneurial exit postponement is best understood as the outcome of structured decision-making under uncertainty combined with biased belief formation. By integrating insights from entrepreneurship, behavioral economics, and finance, this thesis provides a coherent framework for analyzing when delayed exit reflects rational flexibility and when it constitutes inefficient persistence. This perspective opens new avenues for empirical research and offers a more nuanced and comprehensive understanding of one of the most consequential decisions in the entrepreneurial process: the entrepreneurial exit.

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Lilly Gerlach

Berlin, March 3, 2026